

# **Mosqueda v. Pilipino Banana Growers & Exporters Association, Inc. (G.R. Nos. 189185 & 189305, August 16, 2016)**

G.R. No. 189185, August 16, 2016,

&#9830; Decision, Bersamin, [J]

&#9830; Concurring Opinion, Leonen, [J]

[ G.R. No. 189185. August 16, 2016 ]

WILFREDO MOSQUEDA, MARCELO VILLAGANES, JULIETA LAWAGON, CRISPIN ALCOMENDRAS, CORAZON SABINADA, VIRGINIA CATA-AG, FLORENCIA SABANDON, AND LEDEVINA ADLAWAN, PETITIONERS, VS. PILIPINO BANANA GROWERS & EXPORTERS ASSOCIATION, INC., DAVAO FRUITS CORPORATION, AND LAPANDAY AGRICULTURAL AND DEVELOPMENT CORPORATION, RESPONDENTS.

CITY GOVERNMENT OF DAVAO, PETITIONER, VS. COURT OF APPEALS, PILIPINO BANANA GROWERS & EXPORTERS ASSOCIATION (PBGEA), DAVAO FRUITS CORPORATION, AND LAPANDAY AGRICULTURAL AND DEVELOPMENT CORPORATION, RESPONDENTS.

This appeal through the consolidated petitions for review on certiorari assails the decision promulgated on January 9, 2009 whereby the Court of Appeals (CA) reversed and set aside the judgment rendered on September 22, 2007 by the Regional Trial Court (RTC), Branch 17, in Davao City upholding the validity and constitutionality of Davao City Ordinance No. 0309-07, to wit:

WHEREFORE, premises considered, the appeal is GRANTED. The assailed September 22, 2007 Decision of the Regional Trial Court (RTC), 11th Judicial Region, Branch 17, Davao City, upholding the validity and constitutionality of Davao City Ordinance No. 0309-07, is hereby REVERSED and SET ASIDE.

FURTHER, the Writ of Preliminary Injunction dated 28 January 2008 enjoining the City Government of Davao, and any other person or entity acting in its behalf, from enforcing and implementing City Ordinance No. 0309-07, is hereby made permanent.

After several committee hearings and consultations with various stakeholders, the Sangguniang Panlungsod of Davao City enacted Ordinance No. 0309, Series of 2007, to impose a ban against aerial spraying as an agricultural practice by all agricultural entities within Davao City, viz.:

AN ORDINANCE BANNING AERIAL SPRAYING AS AN AGRICULTURAL PRACTICE IN ALL AGRICULTURAL ACTIVITIES BY ALL AGRICULTURAL ENTITIES IN DAVAO CITY

Be it enacted by the Sangguniang Panlungsod of Davao City in session assembled that:

SECTION 1. TITLE. This Ordinance shall be known as "An Ordinance Banning Aerial Spraying as an Agricultural Practice in all Agricultural Activities by all Agricultural Entities in Davao City";

SECTION 2. POLICY OF THE CITY. It shall be the policy of the City of Davao to eliminate the method of aerial spraying as an agricultural practice in all agricultural activities by all entities within Davao City;

SECTION 3. DEFINITION OF TERMS:

a. Aerial Spraying - refers to application of substances through the use of aircraft of any form which dispenses the substances in the air.

b. Agricultural Practices - refer to the practices conducted by agricultural entities in relation to their agricultural activities;

c. Agricultural Activities - refer to activities that include, but not limited to, land preparation, seeding, planting, cultivation, harvesting and bagging;

d. Agricultural Entities - refer to persons, natural or juridical, involved in agricultural activities

e. Buffer Zone - is an identified 30-meter zone within and around the boundaries of agricultural farms/plantations that need special monitoring to avoid or minimize harm to the environment and inhabitants pursuant to policies and guidelines set forth

in this Ordinance and other government regulations. It is an area of land that must lie within the property which does not include public lands, public thoroughfares or adjacent private properties. It must be planted with diversified trees that grow taller than what are usually planted and grown in the plantation to protect those within the adjacent fields, neighboring farms, residential area, schools and workplaces.

SECTION 4. SCOPE AND APPLICABILITY - The provisions of this Ordinance shall apply to all agricultural entities within the territorial jurisdiction of Davao City;

SECTION 5. BAN OF AERIAL SPRAYING - A ban on aerial spraying shall be strictly enforced in the territorial jurisdiction of Davao City three (3) months after the effectivity of this Ordinance.

SECTION 6. BUFFER ZONE - Consistent with national legislation and government regulations, all agricultural entities must provide for a thirty (30) meter buffer zone within the boundaries of their agricultural farms/plantations. This buffer zone must be properly identified through Global Positioning System (GPS) survey. A survey plan showing the metes and bounds of each agricultural farm/plantation must be submitted to the City Mayor's Office, with the buffer zone clearly identified therein;

SECTION 7. PENAL PROVISION - Violation of any provision of this Ordinance shall be punished as follows:

a. First Offense: Fine of ₱5,000.00 and imprisonment of not less than one (1) month but not more than three (3) months;

b. Second Offense: Fine of ₱5,000.00 and imprisonment of not less than three (3) months but not more than six (6) months and suspension of City-issued permits and licenses for one (1) year;

c. Third Offense: Fine of ₱5,000.00 and imprisonment of not less than six (6) months but not more than one (1) year and perpetual cancellation of City issued permits and licenses;

Provided, that in case the violation has been committed by a juridical person, the person in charge of the management thereof shall be held liable;

SECTION 8. REPEALING CLAUSE - Any Ordinance that is contrary to or inconsistent with any of the provisions of this Ordinance shall be deemed amended or repealed accordingly.

SECTION 9. EFFECTIVITY - This Ordinance shall take effect thirty (30) days from its publication in a newspaper of general circulation in Davao City;

ENACTED, January 23, 2007 by a majority vote of all the Members of the Sangguniang Panlungsod.<sup>2</sup>

City Mayor Rodrigo Duterte approved the ordinance on February 9, 2007.<sup>3</sup> The ordinance took effect on March 23, 2007 after its publication in the newspaper Mindanao Pioneer.<sup>4</sup> Pursuant to Section 5 of the ordinance, the ban against aerial spraying would be strictly enforced three months thereafter.

The Pilipino Banana Growers and Exporters Association, Inc. (PBGEA) and two of its members, namely: Davao Fruits Corporation and Lapanday Agricultural and Development Corporation (PBGEA, et al.), filed their petition in the RTC to challenge the constitutionality of the ordinance, and to seek the issuance of provisional reliefs through a temporary restraining order (TRO) and/or writ of preliminary injunction.<sup>5</sup> They alleged that the ordinance exemplified the unreasonable exercise of police power; violated the equal protection clause; amounted to the confiscation of property without due process of law; and lacked publication pursuant] to Section 5116 of Republic Act No. 7160 (Local Government Code).

On May 8, 2007, the residents living within and adjacent to banana plantations in Davao City led by Wilfredo Mosqueda,<sup>7</sup> joined by other residents of Davao City,<sup>8</sup> (Mosqueda, et al.) submitted their Motion for Leave to Intervene and Opposition to the Issuance of a Preliminary Injunction.<sup>9</sup> The RTC granted their motion on June 4, 2007.<sup>10</sup>

On June 20, 2007, the RTC granted the prayer for issuance of the writ of preliminary injunction, and subsequently issued the writ.<sup>11</sup>

On September 22, 2007, after trial, the RTC rendered judgment declaring Ordinance No. 0309-07 valid and constitutional, decreeing thusly:

WHEREFORE, finding the subject [O]rdinance No. 0309-07 valid and constitutional in all aspect of the grounds assailed by the petitioner, said [C]ity [O]rdinance No. 0309-07, is sustained of its validity and constitutionality.

Accordingly, the order of this court dated June 20, 2007, granting the writ of preliminary injunction as prayed for by petitioner is ordered cancelled and set aside as a result of this decision.

The RTC opined that the City of Davao had validly exercised police power<sup>13</sup> under the General Welfare Clause of the Local Government Code;<sup>14</sup> that the ordinance, being based on a valid classification, was consistent with the Equal Protection Clause; that aerial spraying was distinct from other methods of pesticides application because it exposed the residents to a higher degree of health risk caused by aerial drift;<sup>15</sup> and that the ordinance enjoyed the presumption of constitutionality, and could be invalidated only upon a clear showing that it had violated the Constitution.<sup>16</sup>

However, the RTC, recognizing the impracticability of the 3-month transition period under Section 5 of Ordinance No. 0309-07, recommended the parties to agree on an extended transition period.<sup>17</sup>

PBGEA, et al. appealed,<sup>18</sup> and applied for injunctive relief from the CA,<sup>19</sup> which granted the application<sup>20</sup> and consequently issued a TRO to meanwhile enjoin the effectivity of the ordinance.<sup>21</sup>

On January 9, 2009, the CA promulgated its assailed decision reversing the judgment of the RTC.<sup>22</sup> It declared Section 5 of Ordinance No. 0309-07 as void and unconstitutional for being unreasonable and oppressive; found the three-month transition period impractical and oppressive in view of the engineering and technical requirements of switching from aerial spraying to truck-mounted boom spraying; and opined that the ban ran afoul with the Equal Protection Clause inasmuch as Section 3(a) of the ordinance - which defined the term aerial spraying - did not make reasonable distinction between the hazards, safety and beneficial effects of liquid substances that were being applied aerially; the different classes of pesticides or fungicides; and the levels of concentration of these substances that could be beneficial and could enhance agricultural production.

The CA did not see any established relation between the purpose of protecting the public and the environment against the harmful effects of aerial spraying, on one hand, and the imposition of the ban against aerial spraying of all forms of substances, on the other. It ruled that the maintenance of the 30-meter buffer zone within and around the agricultural plantations under Section 6 of Ordinance No. 0309-07 constituted taking of property without due process because the landowners were thereby compelled to cede portions of their property without just compensation; that the exercise of police power to require the buffer zone was invalid because there was no finding that the 30-meter surrounding belt was obnoxious to the public welfare; and that, accordingly, Ordinance No. 0309-07 was unconstitutional because of the absence of a separability clause.

The City of Davao and the intervenors filed their respective motions for reconsideration, but the CA denied the motions on August 7, 2009.<sup>23</sup>

Hence, the separate, but now consolidated, appeals by petition for review *on certiorari*.

In G.R. No. 189185, petitioners Mosqueda, et al. rely on the following grounds, namely:

THE COURT OF APPEALS IGNORED FUNDAMENTAL PRECEPTS AND CONCEPTS OF LAW WHICH, PROPERLY CONSIDERED, NECESSARILY LEAD TO THE CONCLUSION THAT THE DAVAO ORDINANCE IS CONSTITUTIONAL AND VALID

THE DAVAO ORDINANCE IS CONSISTENT WITH THE EQUAL PROTECTION CLAUSE

THE MEANS EMPLOYED BY THE DAVAO ORDINANCE IS MORE THAN REASONABLY RELATED TO THE PURPOSE IT SEEKS TO ACHIEVE

THE DAVAO ORDINANCE IS VALID, BEING DEMONSTRABLY REASONABLE AND FAIR

THE REQUIREMENT RELATING TO THE 30-METER BUFFER ZONE ARE [SIC] CONSISTENT WITH DUE PROCESS OF LAW, BEING A VALID EXERCISE OF POLICE POWER

Mosqueda, et al. state that the CA ignored well-established precepts like the primacy of human rights over property rights and the presumption of validity in favor of the ordinance; that the CA preferred the preservation of the profits of respondents PBGEA, et al. to the residents' right to life, health and ecology,<sup>24</sup> thereby disregarding the benevolent purpose of the ordinance; that the CA assumed the functions of the lawmaker when it set aside the wisdom behind the enactment of the ordinance; that the CA failed to apply the precautionary principle, by which the State was allowed to take positive actions to prevent harm to the environment and to human health despite the lack of scientific certainty; that the CA erred in applying

the "strict scrutiny method" in holding that the ordinance violated the Equal Protection Clause because it only thereby applied in reviewing classifications that affected fundamental rights; that there was nothing wrong with prohibiting aerial spraying per se considering that even the aerial spraying of water produced drift that could affect unwilling neighbors whose constitutional right to a clean and healthy environment might be impinged;<sup>25</sup> that as far as the three-month period was concerned, the CA should have considered that manual spraying could be conducted while the PBGEA, et al. laid down the preparations for the conduct of boom spraying;<sup>26</sup> that "reasonableness" could be more appropriately weighed by balancing the interests of the parties against the protection of basic rights, like the right to life, to health, and to a balanced and healthful ecology;<sup>27</sup> that PBGEA, et al. did not substantiate their claim of potential profit losses that would result from the shift; that business profits should remain inferior and subordinate to their fundamental rights as residents of Davao City, which were the rights that the assailed ordinance has sought to protect;<sup>28</sup> that PBGEA, et al. did not explore other modes of pesticide treatment either as a stop-gap or as a temporary measure while shifting to truck mounted boom spraying;<sup>29</sup> that the imposition of the 30-meter buffer zone was a valid exercise of police power that necessarily flowed from the protection afforded by the ordinance from the unwanted effects of ground spraying; that the imposition of the buffer zone did not constitute compensable taking under police power, pursuant to the pronouncements in *Seng Kee & Co. v. Earnshaw* and *Piatt*<sup>30</sup> *Patalinghug v. Court of Appeals*,<sup>31</sup> and *Social Justice Society (SJS) v. Atienza, Jr.*;<sup>32</sup> and that the 30-meter buffer zone conformed with the ISO 14000<sup>33</sup> and the DENR Environmental Compliance Certificate (ECC) requirement.<sup>34</sup>

In G.R. No. 189305, petitioner City of Davao submits the following as the issues to be considered and resolved, to wit:

WHETHER OR NOT THE HONORABLE COURT OF APPEALS ERRED IN HOLDING THAT SECTION 5 OF ORDINANCE NO. 0309-07, SERIES OF 2007 IS OPPRESSIVE AND AN UNREASONABLE EXERCISE OF DELEGATED POLICE POWER

WHETHER OR NOT THE HONORABLE COURT OF APPEALS ERRED IN HOLDING THAT ORDINANCE NO. 0309-07 IS VIOLATIVE OF THE EQUAL PROTECTION CLAUSE OF THE CONSTITUTION;

WHETHER OR NOT THE HONORABLE COURT OF APPEALS ERRED IN HOLDING THAT ORDINANCE NO. 0309-07 CONSTITUTES TAKING OF PROPERTY WITHOUT COMPENSATION, THUS, VIOLATIVE OF THE DUE PROCESS CLAUSE OF THE CONSTITUTION

WHETHER OR NOT AERIAL SPRAYING OF FUNGICIDES IS SAFE TO THE PEOPLE AND THE ENVIRONMENT

The City of Davao explains that it had the authority to enact the assailed ordinance because it would thereby protect the environment and regulate property and business in the interest of the general welfare pursuant to Section 458 of the Local Government Code;<sup>35</sup> that the ordinance was enacted to carry out its mandate of promoting the public welfare under the General Welfare Clause (Section 16 of the Local Government Code); that the ordinance did not violate the Equal Protection Clause because the distinction lies in aerial spray as a method of application being more deleterious than other modes; that aerial spraying produces more drift that causes discomfort, and an extremely offensive and obnoxious experience the part of the residents; that spray drift cannot be controlled even with use by the respondents of highly advanced apparatus, such as the Differential Global Positioning System, Micronair Rotary Drift Control Atomizers, Intellimap, Intelliflow Spray Valve System, Control and Display Unit and the Target Flow Spray Valve Switch System;<sup>36</sup> that because of the inherent toxicity of Mancozeb (the fungicide aerially applied by the respondents), there is no need to provide for a substantial distinction based on the level of concentration;<sup>37</sup> that as soon as fungicides are released in the air, they become air pollutants pursuant to Section 5 of Republic Act No. 8749 (Philippine Clean Air Act of 1999),<sup>38</sup> and the activity thus falls under the authority of the local government units to ban; and that the ordinance does not only seek to protect and promote human health but also serves as a measure against air pollution.

The City of Davao insists that it validly exercised police power because it does not thereby oblige the shift from aerial to truck-mounted boom spraying; that the respondents only choose boom spraying to justify the alleged impracticability of the transition period by erroneously adding the months required for each of the stages without considering other steps that may be simultaneously undertaken;<sup>39</sup> that the Court should apply its ruling in *Social Justice Society v. Atienza, Jr.*,<sup>40</sup> by which the six-month period for the folding-up of business operations was declared a legitimate exercise of police power; that the respondents did not present any documentary evidence on the feasibility of adopting other methods;<sup>41</sup> that only 1,800 hectares out of 5,200 hectares of plantations owned and operated by PBGEA's members use aerial spraying, hence, the perceived ominous consequence of imposing a ban on aerial spray to the banana industry is entirely misleading;<sup>42</sup> that the urgency of prohibiting aerial spray justifies the three-month transition period; that the complaints of the community residents

- ranging from skin itchiness, contraction and/or tightening in the chest, nausea, appetite loss and difficulty in breathing after exposure to spray mist - only prove that aerial spraying brings discomfort and harm to the residents; that considering that the testimony of Dr. Lynn Crisanta R. Panganiban, a pharmacologist and toxicologist, established that fungicides could cause debilitating effects on the human body once inhaled or digested, the CA erred in holding that there was no correlation between aerial application and the complaints of the residents; that given that aerial spray produces more drift and is uncontrollable compared to the other methods of applying fungicides, the ordinance becomes reasonable;<sup>43</sup> and that the medical-related complaints of the residents need not be proven by medical records considering that these were based on personal knowledge.<sup>44</sup>

The City of Davao contends that the imposition of the 30-meter buffer zone is a valid exercise of police power, rendering the claim for just compensation untenable; that the maintenance of the buffer zone does not require the respondents to cede a portion of their landholdings; that the planting of diversified trees within the buffer zone will serve to insulate the residents from spray drift; that such buffer zone does not deprive the landowners of the lawful and beneficial use of their property;<sup>45</sup> and that the buffer zone is consistent with the Constitution, which reminds property owners that the use of property bears a social function.<sup>46</sup>

In their comment, the respondents posit that the petition of the City of Davao should be dismissed for failure to attach material portions of the records, and for raising factual errors that are not within the realm of this appeal by petition for review on certiorari;<sup>47</sup> that the CA correctly declared the ordinance as unreasonable due to the impossibility of complying with the three-month transition period; that shifting from aerial to truck-mounted boom spraying will take at least three years and entails careful planning, equipment and machineries, civil works, and capital funding of at least ₱400,000,000.00;<sup>48</sup> that the Court could rely on its ruling in *City of Manila v. Laguio, Jr.*,<sup>49</sup> where an ordinance directing an existing establishment to wind up or to transfer its business was declared as confiscatory in nature, and, therefore, unconstitutional;<sup>50</sup> that the total ban against aerial spraying, coupled with the inadequate time to shift to truck-mounted boom spraying, effectively deprives the respondents with an efficient means to control the spread of the Black Sigatoka disease that threatens the banana plantations; that the ordinance will only expose the plantations to the virulent disease that is capable of infecting 60% of the plantations on a single cycle;<sup>51</sup> that compared with other modes of application, aerial spraying is more cost-efficient, safe and accurate; that truck-mounted boom spraying, for instance, requires 80-200 liters of solution per hectare,<sup>53</sup> while manual spraying uses 200-300 liters of solution per hectare; that aerial spraying only requires 30 liters per hectare; that in terms of safety and accuracy, manual spraying is the least safe and accurate,<sup>54</sup> and produces more drift than aerial spraying;<sup>55</sup> that due to the 300-liter solution required, the workers will be more exposed to the solution during manual application and such application will thus be more in conflict with the purpose of the ordinance to prevent human exposure;<sup>56</sup> that the respondents also find the irrigation sprinklers suggested by the City of Davao as wasteful, unsafe and impractical because it cannot provide the needed coverage for application of the solution to effectively control the Black Sigatoka disease; that in contrast, aerial application, coupled with the latest state of the art technology and equipment, ensures accuracy, effectiveness, efficiency and safety compared to the other methods of application; that the respondents vouch for the safety of the fungicides they use by virtue of such fungicides having been registered with the Fertilizer and Pesticide Authority (FPA) and classified as Category IV,<sup>57</sup> and found to be mild; and that oral ingestion in large doses is required before any adverse effects to humans may result.<sup>58</sup>

The respondents lament that the ban was imposed without any scientific basis; that the report<sup>59</sup> prepared by a fact-finding team (composed of the Vice Mayor, the City Health Officer, The City Planning and Development Coordinator and the Assistance City Planning and Development Coordinator) organized by the City of Davao revealed that there was no scientific evidence to support the clamor for the ban against aerial spraying; that furthermore, national government agencies like the Department of Agriculture (DA), Department of Health (DOR) and the Department of Trade and Industry (DTI) similarly concluded that there was no scientific evidence to support the ban;<sup>60</sup> that for four decades since the adoption of aerial spraying, there has been no reported outbreak or any predisposition to ailment connected with the pesticides applied; that the testimonies of the residents during the trial were mere "emotional anecdotal evidence" that did not establish any scientific or medical bases of any causal connection between the alleged health conditions complained of and the fungicides applied during aerial spraying;<sup>61</sup> that the allegations of health and environmental harm brought by the pesticides used to treat the banana plantations were unfounded; that the 2001 study of the International Agency for Research on Cancer showed that, contrary to the claim of Dra. Panganiban, the by-product of Mancozeb (Ethylenethiourea or ETU) was "non-genotoxic" and not expected to produce thyroid cancer;<sup>62</sup> that Carlos Mendoza, a geo-hydrologist and geophysicist,

testified that underground water contamination through aerial spraying would be impossible because of the presence of latex, thick layers of clay and underlying rock formations;<sup>63</sup>that even the study conducted by the Philippine Coconut Authority (PCA) showed that the rhinoceros beetle infestation in coconut plantations adjacent to the banana plantations was due to the farmer's failure to observe phyto-sanitary measures, not to aerial spraying;<sup>64</sup>that furthermore, aerial spraying is internationally accepted as a "Good Agricultural Practice" (GAP)<sup>65</sup>under the International Code of Conduct on the Distribution and Use of Pesticides by the United Nations-Food and Agricultural Organization (UN-FAO); that as such, they observe the standards laid down by the UN-FAO, and utilize aerial spraying equipment that will ensure accuracy, safety and efficiency in applying the substances, and which more than complies with the requirement under the Guidelines on Good Practice for Aerial Application of Pesticides (Rome 2001);<sup>66</sup>that in addition, they strictly observe standard operating procedures prior to take-off,<sup>67</sup>in-flight<sup>68</sup>and post-flight;<sup>69</sup>that they substantially invested in state-of-the-art technology and equipment designed to ensure safety, accuracy, and effectiveness of aerial spraying operations, to avoid aerial drift;<sup>70</sup>that their equipment include: wind meters (to measure the wind velocity in a specific area), wind cones (to determine the wind direction, and whether the wind is a headwind, tailwind or a crosswind); central weather station (to measure wind speed, the temperature and relative humidity), Differential Global Positioning System (DGPS),<sup>71</sup>Intellimap,<sup>72</sup>Control and Display Unit,<sup>73</sup>Micronair Rotary Drift Control Atomizers (AU 5000 Low-Drift model),<sup>74</sup>Intelliflow Spray Valve System,<sup>75</sup>and Target Flow Spray Valve Switch System;<sup>76</sup>and that they want to minimize, if not, eliminate the occurrence of spray drift in order to minimize wastage of resources and reduced efficiency of spraying programs implemented to control the Black Sigatoka disease.<sup>77</sup>

The respondents maintain that Ordinance No. 0309-07 will regulate aerial spraying as a method of application, instead of the substances being used therein; that the prohibition is overbroad in light of other available reasonable measures that may be resorted to by the local government; that the ordinance is unreasonable, unfair, oppressive, and tantamount to a restriction or prohibition of trade;<sup>78</sup>that the ordinance will effectively impose a prohibition against all pesticides, including fungicides that fall under the mildest type of substance; that as such, the petitioner has disregarded existing valid and substantive classifications established and recognized by the World Health Organization (WHO) that are adopted by the FPA; that the FPA is the national agency armed with the professional competence, technical expertise, and legal mandate to deal with the issue of use and application of pesticides in our country; that the fungicides they administer are duly registered with the FPA, and with other more developed countries that have observed a stricter environmental and public health regulation such as the United States Environmental Protection Agency (EPA) and the European Union (EU); that as such, the City of Davao has disregarded valid, substantial and significant distinctions between levels of concentration of the fungicides in the water solution aerially sprayed; that it is the FPA that regulates the level of concentration of agricultural chemicals prior to commercial distribution and use in the country; that the members of PBGEA only spray a water solution (water cocktail) containing 0.1 liter to 1.5 liters of the active ingredient of fungicide in a 30-liter water solution per hectare that has undergone rigorous testing and evaluation prior to registration by the FPA; that the active ingredients of the fungicide are so diluted that no harm may be posed to public health or to the environment through aerial application;<sup>79</sup>that the ordinance was so broad that it prohibits aerial application of any substance, including water;<sup>80</sup>and that aside from fungicides, the respondents also aerially apply vitamins, minerals and organic fertilizers.<sup>81</sup>

The respondents submit that the maintenance of the 30-meter buffer zone under Section 5 of the ordinance constitutes an improper exercise of police power; that the ordinance will require all landholdings to maintain the buffer zone, thereby diminishing to a mere 1,600 square meters of usable and productive land for every hectare of the plantation bounding residential areas, with the zone being reserved for planting "diversified trees;" that this requirement amounts to taking without just compensation or due process; and that the imposition of the buffer zone unduly deprives all landowners within the City of Davao the beneficial use of their property;<sup>82</sup>that the precautionary principle cannot be applied blindly, because its application still requires some scientific basis; that the principle is also based on a mere declaration that has not even reached the level of customary international law, not on a treaty binding on the Government.<sup>83</sup>

The respondents argue that the illegality of the transition period results in the invalidity of the ordinance as it does not carry a separability clause; and that the absence of such clause signifies the intention of the Sangguniang Panlungsod of City of Davao to make the ordinance effective as a whole.<sup>84</sup>

The main issue is whether or not Ordinance No. 0309-07 is unconstitutional on due process and equal protection grounds for being unreasonable and oppressive, and an invalid exercise of police power: (a) in imposing a ban on aerial spraying as an agricultural practice in Davao City under Section 5; (b) in decreeing a 3-month transition-period to shift to other modes of

pesticide application under Section 5; and (c) in requiring the maintenance of the 30-meter buffer zone under Section 6 thereof in all agricultural lands in Davao City.

We deny the petitions for review for their lack of merit.

Preliminary considerations: The significant role of the banana industry in ensuring economic stability and food security

There is no question that the implementation of Ordinance No. 0309-07, although the ordinance concerns the imposition of the ban against aerial spraying in all agricultural lands within Davao City, will inevitably have a considerable impact on the country's banana industry, particularly on export trading.

Banana exportation plays a significant role in the maintenance of the country's economic, stability and food security. Banana is a consistent dollar earner and the fourth largest produced commodity in the Philippines.<sup>85</sup>In 2010, the Philippines figured among the top three banana producing countries in the world.<sup>86</sup>In 2014, fresh bananas accounted for 17% of the country's top agricultural export commodities, gaining a close second to coconut oil with 18%.<sup>87</sup>The Davao Region (Region XI)<sup>88</sup>was the top banana producing region in 2013, with a production growth rate of 16.4%, and 33.76% share in the total agricultural output of the Region.<sup>89</sup>

Despite these optimistic statistics, the banana industry players struggle to keep up with the demands of the trade by combatting the main threat to production posed by two major fungal diseases: the Panama Disease Tropical Race 4 (*Fusarium oxysprum* f.sp. *cubense*) and the Black Sigatoka leaf spot disease (*Mycosphaerella ffiensis* morelet). Pesticides have proven to be effective only against the Black Sigatoka disease. There is yet no known cure for the Panama disease.<sup>90</sup>

The menace of the Black Sigatoka disease cannot be taken lightly. The disease causes destruction of the plant by significantly reducing the leaf area, leading to premature ripening of the produce and resulting in yield losses of at least 50%.<sup>91</sup>Due to its effects on banana export trading, the disease has emerged as a global concern that has correspondingly forced banana producers to increase the use of chemical pesticides.<sup>92</sup>Protectant fungicides such as Mancozeb, chlorothalonil and Propiconazole are applied to combat the disease.<sup>93</sup>These agricultural chemicals are aerially applied by the respondents in the banana plantations within the jurisdiction of Davao City to arrest the proliferation of the disease.

Considering that banana export plantations exist in vast monocultures, effective treatment of the Black Sigatoka disease is done by frequent aerial application of fungicides. This is an expensive practice because it requires permanent landing strips, facilities for the mixing and loading of fungicides, and high recurring expense of spray materials.<sup>94</sup>The cost of aerial spraying accounts to 15-20% of the final retail price of the crop, making the technology essentially unavailable to small landholdings that are more vulnerable to the disease.<sup>95</sup>

Aerial spraying has become an agricultural practice in Davao City since the establishment of the banana plantations in 1960.<sup>96</sup>Out of the 5,205 hectares of commercial plantations devoted to Cavendish banana being operated by the respondents in Davao City,<sup>97</sup>around 1,800 hectares receive treatment through aerial application. These plantations are situated in Barangays Sirib, Manuel Guianga, Tamayong, Subasta Dacudao, Lasang, Mandug, Waan, Tigatto and Callawa,<sup>98</sup>and are affected by the ban imposed by Ordinance No. 0309-07. The DTI has issued a statement to the effect that the ban against aerial spraying in banana plantations "is expected to kill the banana industry," affects the socio-economic development of the barangays hosting the affected plantations, and has a disastrous impact on export trading. The DTI has forecasted that the ban would discourage the entry of new players in the locality, which would have a potential drawback in employment generation.<sup>99</sup>

The Sangguniang Bayan of Davao City enacted Ordinance No. 0309-07 under its corporate powers

The petitioners assert that Ordinance No. 0309-07 is a valid act of the Sangguniang Bayan of Davao City- pursuant to its delegated authority to exercise police power in the furtherance of public welfare and in ensuring a sound and balanced environment for its constituents. The respondents negate this assertion, describing the ordinance as unreasonable, discriminatory and oppressive.

The petitioners' assertion of its authority to enact Ordinance No. 0309-07 is upheld.

To be considered as a valid police power measure, an ordinance must pass a two-pronged test: the formal (i.e., whether the ordinance is enacted within the corporate powers of the local government unit, and whether it is passed in accordance with the procedure prescribed by law); and the substantive (i.e., involving inherent merit, like the conformity of the ordinance with the limitations under the Constitution and the statutes, as well as with the requirements of fairness and reason, and its

consistency with public policy).<sup>100</sup>

The formalities in enacting an ordinance are laid down in Section 53<sup>101</sup> and Section 54<sup>102</sup> of The Local Government Code. These provisions require the ordinance to be passed by the majority of the members of the sanggunian concerned, and to be presented to the mayor for approval. With no issues regarding quorum during its deliberation having been raised, and with its approval of by City Mayor Duterte not being disputed, we see no reason to strike down Ordinance No. 0309-07 for non-compliance with the formal requisites under the Local Government Code.

We next ascertain whether the City of Davao acted within the limits of its corporate powers in enacting Ordinance No. 0309-07.

The corporate powers of the local government unit confer the basic authority to enact legislation that may interfere with personal liberty, property, lawful businesses and occupations in order to promote the general welfare.<sup>103</sup> Such legislative powers spring from the delegation thereof by Congress through either the Local Government Code or a special law. The General Welfare Clause in Section 16 of the Local Government Code embodies the legislative grant that enables the local government unit to effectively accomplish and carry out the declared objects of its creation, and to promote and maintain local autonomy.<sup>104</sup> Section 16 reads:

Sec. 16. General Welfare. Every local government unit shall exercise the powers expressly granted, those necessarily implied therefrom, as well as powers necessary, appropriate, or incidental for its efficient and effective governance, and those which are essential to the promotion of the general welfare. Within their respective territorial jurisdictions, local government units shall ensure and support among other things, the preservation and enrichment of culture, promote health and safety, enhance the right of the people to a balanced ecology, encourage and support the development of appropriate and self-reliant scientific and technological capabilities, improve public morals, enhance economic prosperity and social justice, promote full employment among their residents, maintain peace and order, and preserve the comfort and convenience of their inhabitants.

Section 16 comprehends two branches of delegated powers, namely: the general legislative power and the police power proper. General legislative power refers to the power delegated by Congress to the local legislative body, or the Sangguniang Panlungsod in the case of Dayao City,<sup>105</sup> to enable the local legislative body to enact ordinances and make regulations. This power is limited in that the enacted ordinances must not be repugnant to law, and the power must be exercised to effectuate and discharge the powers and duties legally conferred to the local legislative body. The police power proper, on the other hand, authorizes the local government unit to enact ordinances necessary and proper for the health and safety, prosperity, morals, peace, good order, comfort, and convenience of the local government unit and its constituents, and for the protection of their property.<sup>106</sup>

Section 458 of the Local Government Code explicitly vests the local government unit with the authority to enact legislation aimed at promoting the general welfare, viz.:

Section 458. Powers, Duties, Functions and Compensation. (a) The sangguniang panlungsod, as the legislative body of the city, shall enact ordinances, approve resolutions and appropriate funds for the general welfare of the city and its inhabitants pursuant to Section 16 of this Code and in the proper exercise of the corporate powers of the city as provided for under Section 22 of this Code. x x x

In terms of the right of the citizens to health and to a balanced and healthful ecology, the local government unit takes its cue from Section 15 and Section 16, Article II of the 1987 Constitution. Following the provisions of the Local Government Code and the Constitution, the acts of the local government unit designed to ensure the health and lives of its constituents and to promote a balanced and healthful ecology are well within the corporate powers vested in the local government unit. Accordingly, the Sangguniang Bayan of Davao City is vested with the requisite authority to enact an ordinance that seeks to protect the health and well-being of its constituents.

The respondents pose a challenge against Ordinance No. 0309-07 on the ground that the Sangguniang Bayan of Davao City has disregarded the health of the plantation workers, contending that by imposing the ban against aerial spraying the ordinance would place the plantation workers at a higher health risk because the alternatives of either manual or truck-boom spraying method would be adopted; and that exposing the workers to the same risk sought to be prevented by the ordinance would defeat its purported purpose.

We disagree with the respondents.

With or without the ban against aerial spraying, the health and safety of plantation workers are secured by existing state policies, rules and regulations implemented by the FPA, among others, which the respondents are lawfully bound to comply with. The respondents even manifested their strict compliance with these rules, including those in the UN-FAO Guidelines on Good Practice for Aerial Application of Pesticides (Rome 2001). We should note that the Rome 2001 guidelines require the pesticide applicators to observe the standards provided therein to ensure the health and safety of plantation workers. As such, there cannot be any imbalance between the right to health of the residents vis-a-vis the workers even if a ban will be imposed against aerial spraying and the consequent adoption of other modes of pesticide treatment.

Furthermore, the constitutional right to health and maintaining environmental integrity are privileges that do not only advance the interests of a group of individuals. The benefits of protecting human health and the environment transcend geographical locations and even generations. This is the essence of Sections 15 and 16, Article II of the Constitution. In *Oposa v. Factoran, Jr.*<sup>107</sup> we declared that the right to a balanced and healthful ecology under Section 16 is an issue of transcendental importance with intergenerational implications. It is under this milieu that the questioned ordinance should be appreciated.

Advancing the interests of the residents who are vulnerable to the alleged health risks due to their exposure to pesticide drift justifies the motivation behind the enactment of the ordinance. The City of Davao has the authority to enact pieces of legislation that will promote the general welfare, specifically the health of its constituents. Such authority should not be construed, however, as a valid license for the City of Davao to enact any ordinance it deems fit to discharge its mandate. A thin but well-defined line separates authority to enact legislations from the method of accomplishing the same.

By distinguishing authority from method we face this question: Is a prohibition against aerial spraying a lawfully permissible method that the local government unit of Davao City may adopt to prevent the purported effects of aerial drift? To resolve this question, the Court must dig deeper into the intricate issues arising from these petitions.

Ordinance No. 0309-07 violates the Due Process Clause

A valid ordinance must not only be enacted within the corporate powers of the local government and passed according to the procedure prescribed by law.<sup>108</sup> In order to declare it as a valid piece of local legislation, it must also comply with the following substantive requirements, namely: (1) it must not contravene the Constitution or any statute; (2) it must be fair, not oppressive; (3) it must not be partial or discriminatory; (4) it must not prohibit but may regulate trade; (5) it must be general and consistent with public policy; and (6) it must not be unreasonable.<sup>109</sup>

In the State's exercise of police power, the property rights of individuals may be subjected to restraints and burdens in order to fulfill the objectives of the Government.<sup>110</sup> A local government unit is considered to have properly exercised its police powers only if it satisfies the following requisites, to wit: (1) the interests of the public generally, as distinguished from those of a particular class, require the interference of the State; and (2) the means employed are reasonably necessary for the attainment of the object sought to be accomplished and not unduly oppressive.<sup>111</sup> The first requirement refers to the Equal Protection Clause of the Constitution; the second, to the Due Process Clause of the Constitution.<sup>112</sup>

Substantive due process requires that a valid ordinance must have a sufficient justification for the Government's action.<sup>113</sup> This means that in exercising police power the local government unit must not arbitrarily, whimsically or despotically enact the ordinance regardless of its salutary purpose. So long as the ordinance realistically serves a legitimate public purpose, and it employs means that are reasonably necessary to achieve that purpose without unduly oppressing the individuals regulated, the ordinance must survive a due process challenge.<sup>114</sup>

The respondents challenge Section 5 of Ordinance No. 0309-07 for being unreasonable and oppressive in that it sets the effectivity of the ban at three months after publication of the ordinance. They allege that three months will be inadequate time to shift from aerial to truck-mounted boom spraying, and effectively deprives them of efficient means to combat the Black Sigatoka disease.

The petitioners counter that the period is justified considering the urgency of protecting the health of the residents.

The impossibility of carrying out a shift to another mode of pesticide application within three months can readily be appreciated given the vast area of the affected plantations and the corresponding resources required therefor. To recall, even the RTC recognized the impracticality of attaining a full-shift to other modes of spraying within three months in view of

the costly financial and civil works required for the conversion.<sup>115</sup>In the assailed decision, the CA appropriately observed:

There appears to be three (3) forms of ground spraying, as distinguished from aerial spraying, which are: 1. "Truck-mounted boom spraying;" 2. "manual or backpack spraying." and 3. "sprinkler spraying." Petitioners-appellants claim that it was physically impossible for them to shift to "truck-mounted boom spraying" within three (3) months before the aerial spraying ban is actually enforced. They cited the testimony of Dr. Maria Emilia Rita G. Fabregar, Ph.D, PBGEA Chairperson, to the effect that since banana plantations in Davao City were configured for aerial spraying, the same lack the road network to make "truck-mounted boom spraying" possible. According to Dr. Fabregar, it was impossible to construct such road networks in a span of three (3) months. Engr. Magno P. Porticos, Jr., confirmed that the shift demands the construction of three hundred sixty (360) linear kilometers of road which cannot be completed in three (3) months.

In their separate testimonies, Dr. Fabregar and Engr. Porticos explained that a shift to "truck-mounted boom spraying" requires the following steps which may be completed in three (3) years:

1. six (6) months for planning the reconfiguration of banana plantations to ensure effective truck-mounted boom spraying for the adequate protections of the plantations from the Black Sigatoka fungus and other diseases, while maximizing land use;
2. two (2) months to secure government permits for infrastructure works to be undertaken thereon;
3. clearing banana plants and dismantling or reconstructing fixed infrastructures, such as roads, drains, cable ways, and irrigation facilities, which phase may be completed in eighteen (18) months;
4. importation and purchase of trucks mounted with boom spraying, nurse trucks and protective gears. The placing of orders and delivery of these equipments, including the training [of] the personnel who would man the same, would take six (6) months; and
5. securing the needed capitalization to finance these undertakings would take six (6) months to a year.

Ms. Maria Victoria E. Sembrano, CPA, Chairperson of the PBGEA Finance Committee, testified that her committee and the Technical Committee and Engineering Group of PBGEA conducted a feasibility study to determine the cost in undertaking the shift to ground spraying. Their findings fixed the estimated cost for the purpose at Php 400 Million.

Both appellees failed to rebut the foregoing testimonies with empirical findings to the contrary.

Thus, in view of the infrastructural requirements as methodically explained, We are convinced that it was physically impossible for petitioners-appellants to carry out a carefully planned configuration of vast hectares of banana plantations and be able to actually adopt "truck-mounted boom spraying" within three (3) months. To compel petitioners-appellants to abandon aerial spraying in favor of "manual or backpack spraying" or "sprinkler spraying" within 3 months puts petitioners-appellants in a vicious dilemma between protecting its investments and the health of its workers, on the one hand, and the threat of prosecution if they refuse to comply with the imposition. We even find the 3-months transition period insufficient, not only in acquiring and gearing-up the plantation workers of safety appurtenances, but more importantly in reviewing safety procedures for "manual or backpack spraying" and in training such workers for the purpose. Additionally, the engineering works for a sprinkler system in vast hectares of banana plantations could not possibly be completed within such period, considering that safety and efficiency factors need to be considered in its structural re-designing.

Respondent-appellee argues that the Ordinance merely banned an agricultural practice and did not actually prohibit the operation of banana plantations; hence, it is not oppressive. While We agree that the measure did not impose a closure of a lawful enterprise, the proviso in Section 5, however, compels petitioners-appellants to abandon aerial spraying without affording them enough time to convert and adopt other spraying practices. This would preclude petitioners-appellants from being able to fertilize their plantations with essential vitamins and minerals substances, aside from applying thereon the needed fungicides or pesticides to control, if not eliminate the threat of, plant diseases. Such an apparent eventuality would prejudice the operation of the plantations, and the economic repercussions thereof would just be akin to shutting down the venture.

This Court, therefore, finds Section 5 of Ordinance No. 0309-07 an invalid provision because the compulsion thereunder to abandon aerial spraying within an impracticable period of "three (3) months after the effectivity of this Ordinance" is "unreasonable, oppressive and impossible to comply with."<sup>116</sup>

The required civil works for the conversion to truck-mounted boom spraying alone will consume considerable time and financial resources given the topography and geographical features of the plantations.<sup>117</sup>As such, the conversion could not be completed within the short timeframe of three months. Requiring the respondents and other affected individuals to comply with the consequences of the ban within the three-month period under pain of penalty like fine, imprisonment and even cancellation of business permits would definitely be oppressive as to constitute abuse of police power.

The respondents posit that the requirement of maintaining a buffer zone under Section 6 of the ordinance violates due process for being confiscatory; and that the imposition unduly deprives all agricultural landowners within Davao City of the beneficial use of their property that amounts to taking without just compensation.

The position of the respondents is untenable.

In *City of Manila v. Laguio, Jr.*,<sup>118</sup> we have thoroughly explained that taking only becomes confiscatory if it substantially divests the owner of the beneficial use of its property, viz.:

An ordinance which permanently restricts the use of property that it cannot be used for any reasonable purpose goes beyond regulation and must be recognized as a taking of the property without just compensation. It is intrusive and violative of the private property rights of individuals.

The Constitution expressly provides in Article III, Section 9, that "private property shall not be taken for public use without just compensation." The provision is the most important protection of property rights in the Constitution. This is a restriction on the general power of the government to take property. The constitutional provision is about ensuring that the government does not confiscate the property of some to give it to others. In part too, it is about loss spreading. If the government takes away a person's property to benefit society, then society should pay. The principal purpose of the guarantee is "to bar the Government from forcing some people alone to bear public burdens which, in all fairness and justice, should be borne by the public as a whole.

There are two different types of taking that can be identified. A "possessory" taking occurs when the government confiscates or physically occupies property. A "regulatory" taking occurs when the government's regulation leaves no reasonable economically viable use of the property.

In the landmark case of *Pennsylvania Coal v. Mahon*, it was held that a taking also could be found if government regulation of the use of property went "too far." When regulation reaches a certain magnitude, in most if not in all cases there must be an exercise of eminent domain and compensation to support the act. While property may be regulated to a certain extent, if regulation goes too far it will be recognized as a taking.

No formula or rule can be devised to answer the questions of what is too far and when regulation becomes a taking. In *Mahon*, Justice Holmes recognized that it was "a question of degree and therefore cannot be disposed of by general propositions." On many other occasions as well, the U.S. Supreme Court has said that the issue of when regulation constitutes a taking is a matter of considering the facts in each case. The Court asks whether justice and fairness require that the economic loss caused by public action must be compensated by the government and thus borne by the public as a whole, or whether the loss should remain concentrated on those few persons subject to the public action.

What is crucial in judicial consideration of regulatory takings is that government regulation is a taking if it leaves no reasonable economically viable use of property in a manner that interferes with reasonable expectations for use. A regulation that permanently denies all economically beneficial or productive use of land is, from the owner's point of view, equivalent to a "taking" unless principles of nuisance or property law that existed when the owner acquired the land make the use prohibitable. When the owner of real property has been called upon to sacrifice all economically beneficial uses in the name of the common good, that is, to leave his property economically idle, he has suffered a taking.

A regulation which denies all economically beneficial or productive use of land will require compensation under the takings clause. Where a regulation places limitations on land that fall short of eliminating all economically beneficial use, a taking nonetheless may have occurred, depending on a complex of factors including the regulation's economic effect on the landowner, the extent to which the regulation interferes with reasonable investment-backed expectations and the character of government action. These inquiries are informed by the purpose of the takings clause which is to prevent the government from forcing some people alone to bear public burdens which, in all fairness and justice, should be borne by the public as a whole.

A restriction on use of property may also constitute a "taking" if not reasonably necessary to the effectuation of a substantial public purpose or if it has an unduly harsh impact on the distinct investment-backed expectations of the owner. (bold emphasis supplied)

The establishment of the buffer zone is required for the purpose of minimizing the effects of aerial spraying within and near the plantations. Although Section 3(e) of the ordinance requires the planting of diversified trees within the identified buffer zone, the requirement cannot be construed and deemed as confiscatory requiring payment of just compensation. A landowner may only be entitled to compensation if the taking amounts to a permanent denial of all economically beneficial or productive uses of the land. The respondents cannot be said to be permanently and completely deprived of their landholdings because they can still cultivate or make other productive uses of the areas to be identified as the buffer zones.

Ordinance No. 0309-07 violates the Equal Protection Clause

A serious challenge being posed against Ordinance No. 0309-07 rests on its supposed collision with the Equal Protection Clause. The respondents submit that the ordinance transgresses this constitutional guaranty on two counts, to wit: (1) by prohibiting aerial spraying *per se*, regardless of the substance or the level of concentration of the chemicals to be applied; and (2) by imposing the 30-meter buffer zone in all agricultural lands in Davao City regardless of the sizes of the landholding.

The constitutional right to equal protection requires that all persons or things similarly situated should be treated alike, both as to rights conferred and responsibilities imposed. It requires public bodies and institutions to treat similarly situated individuals in a similar manner. The guaranty of equal protection secures every person within the State's jurisdiction against intentional and arbitrary discrimination, whether occasioned by the express terms of a statute or by its improper execution through the State's duly constituted authorities. The concept of equal justice under the law demands that the State governs impartially, and not to draw distinctions between individuals solely on differences that are irrelevant to the legitimate governmental objective.<sup>119</sup>

Equal treatment neither requires universal application of laws to all persons or things without distinction,<sup>120</sup> nor intends to prohibit legislation by limiting the object to which it is directed or by the territory in which it is to operate.<sup>121</sup> The guaranty of equal protection envisions equality among equals determined according to a valid classification.<sup>122</sup> If the groupings are characterized by substantial distinctions that make real differences, one class may be treated and regulated differently from another.<sup>123</sup> In other words, a valid classification must be: (1) based on substantial distinctions; (2) germane to the purposes of the law; (3) not limited to existing conditions only; and (4) equally applicable to all members of the class.<sup>124</sup>

Based on these parameters, we find for the respondents.

The reasonability of a distinction and sufficiency of the justification given by the Government for its conduct is gauged by using the means-end test.<sup>125</sup> This test requires analysis of: (1) the interests of the public that generally require its exercise, as distinguished from those of a particular class; and (2) the means employed that are reasonably necessary for the accomplishment of the purpose and are not unduly oppressive upon individuals.<sup>126</sup> To determine the propriety of the classification, courts resort to three levels of scrutiny, viz: the rational scrutiny, intermediate scrutiny and strict scrutiny.

The rational basis scrutiny (also known as the rational relation test or rational basis test) demands that the classification reasonably relate to the legislative purpose.<sup>127</sup> The rational basis test often applies in cases involving economics or social welfare,<sup>128</sup> or to any other case not involving a suspect class.<sup>129</sup>

When the classification puts a quasi-suspect class at a disadvantage, it will be treated under intermediate or heightened review. Classifications based on gender or illegitimacy receive intermediate scrutiny.<sup>130</sup> To survive intermediate scrutiny, the law must not only further an important governmental interest and be substantially related to that interest, but the justification for the classification must be genuine and must not depend on broad generalizations.<sup>131</sup>

The strict scrutiny review applies when a legislative classification impermissibly interferes with the exercise of a fundamental right or operates to the peculiar class disadvantage of a suspect class. The Government carries the burden to prove that the classification is necessary to achieve a compelling state interest, and that it is the least restrictive means to protect such interest.<sup>132</sup>

The petitioners advocate the rational basis test. In particular, the petitioning residents of Davao City argue that the CA erroneously applied the strict scrutiny approach when it declared that the ordinance violated the Equal Protection Clause

because the ban included all substances including water and vitamins. The respondents agree with the CA, however, and add that the ordinance does not rest on a valid distinction because it has lacked scientific basis and has ignored the classifications of pesticides observed by the FPA.

We partly agree with both parties.

In our view, the petitioners correctly argue that the rational basis approach appropriately applies herein. Under the rational basis test, we shall: (1) discern the reasonable relationship between the means and the purpose of the ordinance; and (2) examine whether the means or the prohibition against aerial spraying is based on a substantial or reasonable distinction. A reasonable classification includes all persons or things similarly situated with respect to the purpose of the law.<sup>133</sup>

Applying the test, the established classification under Ordinance No. 0309-07 is to be viewed in relation to the group of individuals similarly situated with respect to the avowed purpose. This gives rise to two classes, namely: (1) the classification under Ordinance No. 0309-07 (legislative classification); and (2) the classification based on purpose (elimination of the mischief). The legislative classification found in Section 4 of the ordinance refers to "all agricultural entities" within Davao City. Meanwhile, the classification based on the purpose of the ordinance cannot be easily discerned because the ordinance does not make any express or implied reference to it. We have to search the voluminous records of this case to divine the animus behind the action of the Sangguniang Panglungsod in prohibiting aerial spraying as an agricultural activity. The effort has led us to the following proposed resolution of the Sangguniang Panglungsod,<sup>134</sup> viz.:

RESOLUTION NO. 1a&#x0448;phi1 \_\_\_\_\_

A RESOLUTION TO ENACT AN ORDINANCE BANNING AERIAL SPRAYING AS AN AGRICULTURAL PRACTICE IN ALL AGRICULTURAL ENTITIES IN DAVAO CITY

WHEREAS, the City of Davao, with fertile lands and ideal climactic condition, hosts various large farms planted with different crops;

WHEREAS, these farms, lay adjacent to other agricultural businesses and that residential areas abuts these farm boundaries;

WHEREAS, aerial spraying as a mode of applying chemical substances such as fungicides and pesticides is being used by investors/companies over large agricultural plantations in Davao City;

WHEREAS, the Davao City watersheds and ground water sources, located within and adjacent to Mount Apo may be affected by the aerial spraying of chemical substances on the agricultural farms and plantations therein;

WHEREAS, the effects of aerial spraying are found to be detrimental to the health of the residents of Davao City most especially the inhabitants nearby agricultural plantations practicing aerials spraying;

WHEREAS, the unstable wind direction during the conduct of aerial spray application of these chemical substances pose health hazards to people, animals, other crops and ground water sources;

WHEREAS, in order to achieve sustainable development, politics must be based on the Precautionary Principle. Environment measures must anticipate, prevent, and attack the causes of environmental degradation. Where there are threats of serious, irreversible damage, lack of scientific certainty should not be used as a reason for postponing measures to prevent environmental degradation;

WHEREAS, it is the policy of the City of Davao to ensure the safety of its inhabitants from all forms of hazards, especially if such hazards come from development activities that are supposed to be beneficial to everybody;

WHEREAS, pesticides are by its nature poisonous, it is all the more dangerous when dispensed aerially through aircraft because of unstable wind conditions which in turn makes aerial spray drifting to unintended targets a commonplace.

WHEREAS, aerial spraying of pesticides is undeniably a nuisance.

WHEREAS, looking at the plight of the complainants and other stakeholders opposed to aerial spraying, the issue of aerial spraying of pesticides is in all fours a nuisance. Given the vastness of the reach of aerial spraying, the said form of dispensation falls into the category of a public nuisance. Public nuisance is defined by the New Civil Code as one which affects a community or neighborhood or any considerable number of persons, although the extent of the annoyance, danger or damage upon individuals may be unequal.

WHEREAS, the General Welfare Clause of the Local Government Code empowers Local Government Units to enact ordinances that provide for the health and safety, promote the comfort and convenience of the City and the inhabitants thereof.

NOW THEREFORE, BE IT RESOLVED AS IT IS HEREBY RESOLVED, that for the health, safety and peace of mind of all the inhabitants of Davao City, let an ordinance be enacted banning aerial spraying as an agricultural practice in all agricultural entities in Davao City.

The proposed resolution identified aerial spraying of pesticides as a nuisance because of the unstable wind direction during the aerial application, which (1) could potentially contaminate the Davao City watersheds and ground water sources; (2) was detrimental to the health of Davao City residents, most especially those living in the nearby plantations; and (3) posed a hazard to animals and other crops. Plainly, the mischief that the prohibition sought to address was the fungicide drift resulting from the aerial application; hence, the classification based on the intent of the proposed ordinance covered all agricultural entities conducting aerial spraying of fungicides that caused drift.

The assailed ordinance thus becomes riddled with several distinction issues.

A brief discussion on the occurrence of the drift that the ordinance seeks to address is necessary.

Pesticide treatment is based on the use of different methods of application and equipment,<sup>135</sup> the choice of which methods depend largely on the objective of distributing the correct dose to a defined target with the minimum of wastage due to "drift."<sup>136</sup> The term "drift" refers to the movement of airborne spray droplets, vapors, or dust particles away from the target area during pesticide application.<sup>137</sup> Inevitably, any method of application causes drift, which may either be primary or secondary. As fittingly described by scholars:<sup>138</sup>

Primary drift is the off-site movement of spray droplets at, or very close to, the time of application. For example, a field application using a boom in a gusty wind situation could easily lead to a primary drift. Primary spray drift is not product specific, and the active ingredients do not differ in their potential to drift. However, the type of formulation, surfactant, or other adjuvant may affect spray drift potential.

Secondary drift is associated with pesticide vapor. Pesticide vapor drift is the movement of the gas that forms when an active ingredient evaporates from plants, soil, or other surfaces. And while vapor drift is an important issue, it only pertains to certain volatile products. Vapor drift and other forms of secondary drift are product specific. Water-based sprays will volatilize more quickly than oil-based sprays. However, oil-based sprays can drift farther, especially above 95F, because they are lighter.

Understandably, aerial drift occurs using any method of application, be it through airplanes, ground sprayers, airblast sprayers or irrigation systems.<sup>139</sup> Several factors contribute to the occurrence of drift depending on the method of application, viz.:

AERIAL  
AIRBLAST  
GROUND  
CHEMIGATION

Droplet size  
Crop canopy  
Droplet size  
Application height

Application height  
Droplet size  
Boom height  
Wind speed

Wind speed  
Wind speed  
Wind speed

The four most common pesticide treatment methods adopted in Davao City are aerial, truck-mounted boom, truck-mounted mechanical, and manual spraying.<sup>140</sup> However, Ordinance No. 0309-07 imposes the prohibition only against aerial spraying.

Davao City justifies the prohibition against aerial spraying by insisting that the occurrence of drift causes inconvenience and harm to the residents and degrades the environment. Given this justification, does the ordinance satisfy the requirement that the classification must rest on substantial distinction?

The occurrence of pesticide drift is not limited to aerial spraying but results from the conduct of any mode of pesticide application. Even manual spraying or truck-mounted boom spraying produces drift that may bring about the same inconvenience, discomfort and alleged health risks to the community and to the environment.<sup>141</sup> A ban against aerial spraying does not weed out the harm that the ordinance seeks to achieve.<sup>142</sup> In the process, the ordinance suffers from being "underinclusive" because the classification does not include all individuals tainted with the same mischief that the law seeks to eliminate.<sup>143</sup> A classification that is drastically underinclusive with respect to the purpose or end appears as an irrational means to the legislative end because it poorly serves the intended purpose of the law.<sup>144</sup>

The claim that aerial spraying produces more aerial drift cannot likewise be sustained in view of the petitioners' failure to substantiate the same. The respondents have refuted this claim, and have maintained that on the contrary, manual spraying produces more drift than aerial treatment.<sup>145</sup> As such, the decision of prohibiting only aerial spraying is tainted with arbitrariness.

Aside from its being underinclusive, the assailed ordinance also tends to be "overinclusive" because its impending implementation will affect groups that have no relation to the accomplishment of the legislative purpose. Its implementation will unnecessarily impose a burden on a wider range of individuals than those included in the intended class based on the purpose of the law.<sup>146</sup>

It can be noted that the imposition of the ban is too broad because the ordinance applies irrespective of the substance to be aerially applied and irrespective of the agricultural activity to be conducted. The respondents admit that they aerially treat their plantations not only with pesticides but also vitamins and other substances. The imposition of the ban against aerial spraying of substances other than fungicides and regardless of the agricultural activity being performed becomes unreasonable inasmuch as it patently bears no relation to the purported inconvenience, discomfort, health risk and environmental danger which the ordinance, seeks to address. The burden now will become more onerous to various entities including the respondents and even others with no connection whatsoever to the intended purpose of the ordinance.

In this respect, the CA correctly observed:

Ordinance No. 0309-07 defines "aerial spraying" as the "application of substances through the use of aircraft of any form which dispenses the substances in the air." Inevitably, the ban imposed therein encompasses aerial application of practically all substances, not only pesticides or fungicides but including water and all forms of chemicals, regardless of its elements, composition, or degree of safety.

Going along with respondent-appellee's ratiocination that the prohibition in the Ordinance refers to aerial spraying as a method of spraying pesticides or fungicides, there appears to be a need to single out pesticides or fungicides in imposing such a ban because there is a striking distinction between such chemicals and other substances (including water), particularly with respect to its safety implications to the public welfare and ecology.

We are, therefore, convinced that the total ban on aerial spraying runs afoul with the equal protection clause because it does not classify which substances are prohibited from being applied aerially even as reasonable distinctions should be made in terms of the hazards, safety or beneficial effects of liquid substances to the public health, livelihood and the environment.<sup>147</sup>

We clarify that the CA did not thereby apply the strict scrutiny approach but only evaluated the classification established by the ordinance in relation to the purpose. This is the essence of the rational basis approach.

The petitioners should be made aware that the rational basis scrutiny is not based on a simple means-purpose correlation; nor does the rational basis scrutiny automatically result in a presumption of validity of the ordinance or deference to the wisdom of the local legislature.<sup>148</sup> To reiterate, aside from ascertaining that the means and purpose of the ordinance are reasonably related, the classification should be based on a substantial distinction.

However, we do not subscribe to the respondents' position that there must be a distinction based on the level of concentration or the classification imposed by the FPA on pesticides. This strenuous requirement cannot be expected from a local government unit that should only be concerned with general policies in local administration and should not be restricted by technical concerns that are best left to agencies vested with the appropriate special competencies. The disregard of the pesticide classification is not an equal protection issue but is more relevant in another aspect of delegated police power that we consider to be more appropriate in a later discussion.

The overinclusiveness of Ordinance No. 0309-07 may also be traced to its Section 6 by virtue of its requirement for the maintenance of the 30- meter buffer zone. This requirement applies regardless of the area of the agricultural landholding, geographical location, topography, crops grown and other distinguishing characteristics that ideally should bear a reasonable relation to the evil sought to be avoided. As earlier discussed, only large banana plantations could rely on aerial technology because of the financial capital required therefor.

The establishment and maintenance of the buffer zone will become more burdensome to the small agricultural landholders because: (1) they have to reserve the 30-meter belt surrounding their property; (2) that will have to be identified through GPS; (3) the metes and bounds of the buffer zone will have to be plotted in a survey plan for submission to the local government unit; and (4) will be limited as to the crops that may be cultivated therein based on the mandate that the zone shall be devoted to "diversified trees" taller than what are being grown therein.<sup>149</sup>The arbitrariness of Section 6 all the more becomes evident when the land is presently devoted to the cultivation of root crops and vegetables, and trees or plants slightly taller than the root crops and vegetables are then to be planted. It is seriously to be doubted whether such circumstance will prevent the occurrence of the drift to the nearby residential areas.

Section 6 also subjects to the 30-meter buffer zone requirement agricultural entities engaging in organic farming, and' do not contribute to the occurrence of pesticide drift. The classification indisputably becomes arbitrary and whimsical.

A substantially overinclusive or underinclusive classification tends to undercut the governmental claim that the classification serves legitimate political ends.<sup>150</sup>Where overinclusiveness is the problem, the vice is that the law has a greater discriminatory or burdensome effect than necessary.<sup>151</sup>In this light, we strike down Section 5 and Section 6 of Ordinance No. 0309-07 for carrying an invidious classification, and for thereby violating the Equal Protection Clause.

The discriminatory nature of the ordinance can be seen from its policy as stated in its Section 2, to wit:

Section 2. POLICY OF THE CITY. It shall be the policy of the City of Davao to eliminate the method of aerial spraying as an agricultural practice in all agricultural activities by all entities within Davao City.

Evidently, the ordinance discriminates against large farmholdings that are the only ideal venues for the investment of machineries and equipment capable of aerial spraying. It effectively denies the affected individuals the technology aimed at efficient and cost-effective operations and cultivation not only of banana but of other crops as well. The prohibition against aerial spraying will seriously hamper the operations of the banana plantations that depend on aerial technology to arrest the spread of the Black Sigatoka disease and other menaces that threaten their production and harvest. As earlier shown, the effect of the ban will not be limited to Davao City in view of the significant contribution of banana export trading to the country's economy.

The discriminatory character of the ordinance makes it oppressive and unreasonable in light of the existence and availability of more permissible and practical alternatives that will not overburden the respondents and those dependent on their operations as well as those who stand to be affected by the ordinance. In the view of Regional Director Roger C. Chio of DA Regional Field Unit XI, the alleged harm caused by aerial spraying may be addressed by following the GAP that the DA has been promoting among plantation operators. He explained his view thusly:

The allegation that aerial spraying is hazardous to animal and human being remains an allegation and assumptions until otherwise scientifically proven by concerned authorities and agencies. This issue can be addressed by following Good Agricultural Practices, which DA is promoting among fruit and vegetable growers/plantations. Any method of agri-chemical application whether aerial or non-aerial if not properly done in accordance with established procedures and code of good agricultural practices and if the chemical applicators and or handlers lack of necessary competency, certainly it could be hazardous. For the assurance that commercial applicators/aerial applicators possessed the competency and responsibility of handling agri-chemical, such applicators are required under Article III, Paragraph 2 of FPA Rules and Regulation No. 1 to secure license from FPA.

Furthermore users and applicators of agri-chemicals are also guided by Section 6 Paragraph 2 and 3 under column of Pesticides and Other agricultural Chemicals of PD 11445 which stated: "FPA shall establish and enforce tolerance levels and good agricultural practices in raw agricultural commodities; to restrict or ban the use of any chemical or the formulation of certain pesticides in specific areas or during certain period upon evidence that the pesticide is eminent [sic] hazards has caused, or is causing widespread serious damage to crops, fish, livestock or to public health and environment."

Besides the aforecited policy, rules and regulation enforced by DA, there are other laws and regulations protecting and preserving the environment. If the implementation and monitoring of all these laws and regulation are closely coordinated with concerned LGUs, Gas and NGAs and other private sectors, perhaps we can maintain a sound and health environment x x x.<sup>152</sup>

Indeed, based on the Summary Report on the Assessment and Factfinding Activities on the Issue of Aerial Spraying in Banana Plantations,<sup>153</sup> submitted by the fact-finding team organized by Davao City, only three out of the 13 barangays consulted by the fact-finding team opposed the conduct of aerial spraying; and of the three barangays, aerial spraying was conducted only in Barangay Subasta. In fact, the fact-finding team found that the residents in those barangays were generally in favor of the operations of the banana plantations, and did not oppose the conduct of aerial spraying.

The Precautionary Principle still requires scientific basis

The petitioners finally plead that the Court should look at the merits of the ordinance based on the precautionary principle. They argue that under the precautionary principle, the City of Davao is justified in enacting Ordinance No. 0309-07 in order to prevent harm to the environment and human health despite the lack of scientific certainty.

The petitioners' plea and argument cannot be sustained.

The principle of precaution originated as a social planning principle in Germany. In the 1980s, the Federal Republic of Germany used the Vorsorgeprinzip ("foresight principle") to justify the implementation of vigorous policies to tackle acid rain, global warming and pollution of the North Sea.<sup>154</sup> It has since emerged from a need to protect humans and the environment from increasingly unpredictable, uncertain, and unquantifiable but possibly catastrophic risks such as those associated with Genetically Modified Organisms and climate change,<sup>155</sup> among others. The oft-cited Principle 15 of the 1992 Rio Declaration on Environment and Development (1992 Rio Agenda), first embodied this principle, as follows:

In order to protect the environment, the precautionary approach shall be widely applied by States according to their capabilities. Where there are threats of serious or irreversible damage, lack of full scientific certainty shall not be used as a reason for postponing cost-effective measures to prevent environmental degradation.

In this jurisdiction, the principle of precaution appearing in the Rules of Procedure for Environmental Cases (A.M. No. 09-6-8-SC) involves matters of evidence in cases where there is lack of full scientific certainty in establishing a causal link between human activity and environmental effect.<sup>156</sup> In such an event, the courts may construe a set of facts as warranting either judicial action or inaction with the goal of preserving and protecting the environment.<sup>157</sup>

It is notable, therefore, that the precautionary principle shall only be relevant if there is concurrence of three elements, namely: uncertainty, threat of environmental damage and serious or irreversible harm. In situations where the threat is relatively certain, or that the causal link between an action and environmental damage can be established, or the probability of occurrence can be calculated, only preventive, not precautionary measures, may be taken. Neither will the precautionary principle apply if there is no indication of a threat of environmental harm; or if the threatened harm is trivial or easily reversible.<sup>158</sup>

We cannot see the presence of all the elements. To begin with, there has been no scientific study. Although the precautionary principle allows lack of full scientific certainty in establishing a connection between the serious or irreversible harm and the human activity, its application is still premised on empirical studies. Scientific analysis is still a necessary basis for effective policy choices under the precautionary principle.<sup>159</sup>

Precaution is a risk management principle invoked after scientific inquiry takes place. This scientific stage is often considered synonymous with risk assessment.<sup>160</sup> As such, resort to the principle shall not be based on anxiety or emotion, but from a rational decision rule, based in ethics.<sup>161</sup> As much as possible, a complete and objective scientific evaluation of the risk to the environment or health should be conducted and made available to decision-makers for them to choose the most appropriate course of action.<sup>162</sup> Furthermore, the positive and negative effects of an activity is also important in the

application of the principle. The potential harm resulting from certain activities should always be judged in view of the potential benefits they offer, while the positive and negative effects of potential precautionary measures should be considered.<sup>163</sup>

The only study conducted to validate the effects of aerial spraying appears to be the Summary Report on the Assessment and Fact-Finding Activities on the Issue of Aerial Spraying in Banana Plantations.<sup>164</sup> Yet, the fact-finding team that generated the report was not a scientific study that could justify the resort to the precautionary principle. In fact, the Sangguniang Bayan ignored the findings and conclusions of the fact-finding team that recommended only a regulation, not a ban, against aerial spraying. The recommendation was in line with the advocacy of judicious handling and application of chemical pesticides by the DOH-Center for Health Development in the Davao Region in view of the scarcity of scientific studies to support the ban against aerial spraying.<sup>165</sup>

We should not apply the precautionary approach in sustaining the ban against aerial spraying if little or nothing is known of the exact or potential dangers that aerial spraying may bring to the health of the residents within and near the plantations and to the integrity and balance of the environment. It is dangerous to quickly presume that the effects of aerial spraying would be adverse even in the absence of evidence. Accordingly, for lack of scientific data supporting a ban on aerial spraying, Ordinance No. 0309-07 should be struck down for being unreasonable.

Ordinance No. 0309-07 is an ultra vires act

The Court further holds that in addition to its unconstitutionality for carrying an unwarranted classification that contravenes the Equal Protection Clause, Ordinance No. 0309-07 suffers from another legal infirmity.

The petitioners represent that Ordinance No. 0309-07 is a valid exercise of legislative and police powers by the Sangguniang Bayan of Davao City pursuant to Section 458 in relation to Section 16 both of the Local Government Code. The respondents counter that Davao City thereby disregarded the regulations implemented by the Fertilizer and Pesticide Authority (FPA), including its identification and classification of safe pesticides and other agricultural chemicals.

An ordinance enjoys the presumption of validity on the basis that:

The action of the elected representatives of the people cannot be lightly set aside. The councilors must, in the very nature of things, be familiar with the necessities of their particular municipality and with all the facts and circumstances which surround the subject, and necessities of their particular municipality and with all the facts and circumstances which surround the subject, and necessitate action. The local legislative body, by enacting the ordinance, has in effect given notice that the regulations are essential to the well-being of the people.<sup>166</sup>

Section 5(c) of the Local Government Code accords a liberal interpretation to its general welfare provisions. The policy of liberal construction is consistent with the spirit of local autonomy that endows local government units with sufficient power and discretion to accelerate their economic development and uplift the quality of life for their constituents.

Verily, the Court has championed the cause of public welfare on several occasions. In so doing, it has accorded liberality to the general welfare provisions of the Local Government Code by upholding the validity of local ordinances enacted for the common good. For instance, in *Social Justice Society (SJS) v. Atienza, Jr.*,<sup>167</sup> the Court validated a zoning ordinance that reclassified areas covered by a large oil depot from industrial to commercial in order to ensure the life, health and property of the inhabitants residing within the periphery of the oil depot. Another instance is *Gancayco v. City Government of Quezon City*,<sup>168</sup> where the Court declared as valid a city ordinance ordering the construction of arcades that would ensure the health and safety of the city and its inhabitants, improvement of their morals, peace, good order, comfort and convenience, as well as the promotion of their prosperity. Even in its early years, the Court already extended liberality towards the exercise by the local government units; of their legislative powers in order to promote the general welfare of their communities. This was exemplified in *United States v. Salaveria*,<sup>169</sup> wherein gambling was characterized as "an act beyond the pale of good morals" that the local legislative council could validly suppress to protect the well-being of its constituents; and in *United States v. Abendan*,<sup>170</sup> whereby the right of the then Municipality of Cebu to enact an ordinance relating to sanitation and public health was upheld.

The power to legislate under the General Welfare Clause is not meant to be an invincible authority. In fact, *Salaveria* and *Abendan* emphasized the reasonableness and consistency of the exercise by the local government units with the laws or policies of the State.<sup>171</sup> More importantly, because the police power of the local government units flows

from the express delegation of the power by Congress, its exercise is to be construed in strictissimi juris. Any doubt or ambiguity arising out of the terms used in granting the power should be construed against the local legislative units.<sup>172</sup> Judicial scrutiny comes into play whenever the exercise of police power affects life, liberty or property.<sup>173</sup> The presumption of validity and the policy of liberality are not restraints on the power of judicial review in the face of questions about whether an ordinance conforms with the Constitution, the laws or public policy, or if it is unreasonable, oppressive, partial, discriminating or in derogation of a common right. The ordinance must pass the test of constitutionality and the test of consistency with the prevailing laws.<sup>174</sup>

Although the Local Government Code vests the municipal corporations with sufficient power to govern themselves and manage their affairs and activities, they definitely have no right to enact ordinances dissonant with the State's laws and policy. The Local Government Code has been fashioned to delineate the specific parameters and limitations to guide each local government unit in exercising its delegated powers with the view of making the local government unit a fully functioning subdivision of the State within the constitutional and statutory restraints.<sup>175</sup> The Local Government Code is not intended to vest in the local government unit the blanket authority to legislate upon any subject that it finds proper to legislate upon in the guise of serving the common good.

The function of pesticides control, regulation and development is within the jurisdiction of the FPA under Presidential Decree No. 1144.<sup>176</sup> The FPA was established in recognition of the need for a technically oriented government entity<sup>177</sup> that will protect the public from the risks inherent in the use of pesticides.<sup>178</sup> To perform its mandate, it was given under Section 6 of Presidential Decree No. 1144 the following powers and functions with respect to pesticides and other agricultural chemicals, viz.:

Section 6. Powers and functions. The FPA shall have jurisdiction, on over all existing handlers of pesticides, fertilizers and other agricultural chemical inputs. The FPA shall have the following powers and functions:

### III. Pesticides and Other Agricultural Chemicals

1. To determine specific uses or manners of use for each pesticide or pesticide formulation;
2. To establish and enforce levels and good agricultural practices for use of pesticides in raw agricultural commodities;
3. To restrict or ban the use of any pesticide or the formulation of certain pesticides in specific areas or during certain periods upon evidence that the pesticide is an imminent hazard, has caused, or is causing widespread serious damage to crops, fish or livestock, or to public health and environment;
5. To inspect the establishment and premises of pesticide handlers to insure that industrial health and safety rules and anti-pollution regulations are followed;
6. To enter and inspect farmers' fields to ensure that only the recommended pesticides are used in specific crops in accordance with good agricultural practice;

Evidently, the FPA was responsible for ensuring the compatibility between the usage and the application of pesticides in agricultural activities and the demands for human health and environmental safety. This responsibility includes not only the identification of safe and unsafe pesticides, but also the prescription of the safe modes of application in keeping with the standard of good agricultural practices.

On the other hand, the enumerated devolved functions to the local government units do not include the regulation and control of pesticides and other agricultural chemicals.<sup>179</sup> The non-inclusion should preclude the Sangguniang Bayan of Davao City from enacting Ordinance No. 0309-07, for otherwise it would be arrogating unto itself the authority to prohibit the aerial application of pesticides in derogation of the authority expressly vested in the FPA by Presidential Decree No. 1144.

In enacting Ordinance No. 0309-07 without the inherent and explicit authority to do so, the City of Davao performed an ultra vires act. As a local government unit, the City of Davao could act only as an agent of Congress, and its every act should always conform to and reflect the will of its principal.<sup>180</sup> As clarified in *Batangas CATV, Inc. v. Court of Appeals*:<sup>181</sup>

[W]here the state legislature has made provision for the regulation of conduct, it has manifested its intention that the subject matter shall be fully covered by the statute, and that a municipality, under its general powers, cannot regulate the same conduct. In *Keller vs. State*, it was held that: "Where there is no express power in the charter of a municipality authorizing it to adopt ordinances regulating certain matters which are specifically covered by a general statute, a municipal ordinance,

insofar as it attempts to regulate the subject which is completely covered by a general statute of the legislature, may be rendered invalid. x x x Where the subject is of statewide concern, and the legislature has appropriated the field and declared the rule, its declaration is binding throughout the State." A reason advanced for this view is that such ordinances are in excess of the powers granted to the municipal corporation.

Since E.O. No. 205, a general law, mandates that the regulation of CATV operations shall be exercised by the NTC, an LGU cannot enact an ordinance or approve a resolution in violation of the said law.

It is a fundamental principle that municipal ordinances are inferior in status and subordinate to the laws of the state. An ordinance in conflict with a state law of general character and statewide application is universally held to be invalid. The principle is frequently expressed in the declaration that municipal authorities, under a general grant of power, cannot adopt ordinances which infringe the spirit of a state law or repugnant to the general policy of the state. In every power to pass ordinances given to a municipality, there is an implied restriction that the ordinances shall be consistent with the general law.<sup>182</sup>(Emphasis ours)

For sure, every local government unit only derives its legislative authority from Congress. In no instance can the local government unit rise above its source of authority. As such, its ordinance cannot run against or contravene existing laws, precisely because its authority is only by virtue of the valid delegation from Congress. As emphasized in *City of Manila v. Laguio, Jr.*:<sup>183</sup>

The requirement that the enactment must not violate existing law gives stress to the precept that local government units are able to legislate only by virtue of their derivative legislative power, a delegation of legislative power from the national legislature. The delegate cannot be superior to the principal or exercise powers higher than those of the latter.

This relationship between the national legislature and the local government units has not been enfeebled by the new provisions in the Constitution strengthening the policy of local autonomy. The national legislature is still the principal of the local government units, which cannot defy its will or modify or violate it.<sup>184</sup>

Moreover, Ordinance No. 0309-07 proposes to prohibit an activity already covered by the jurisdiction of the FPA, which has issued its own regulations under its Memorandum Circular No. 02, Series of 2009, entitled *Good Agricultural Practices for Aerial Spraying of Fungicide in Banana Plantations*.<sup>185</sup> While Ordinance No. 0309-07 prohibits aerial spraying in banana plantations within the City of Davao, Memorandum Circular No. 02 seeks to regulate the conduct of aerial spraying in banana plantations<sup>186</sup> pursuant to Section 6, Presidential Decree No. 1144, and in conformity with the standard of Good Agricultural Practices (GAP). Memorandum Circular No. 02 covers safety procedures,<sup>187</sup> handling<sup>188</sup> and post-application,<sup>189</sup> including the qualifications of applicators,<sup>190</sup> storing of fungicides,<sup>191</sup> safety and equipment of plantation personnel,<sup>192</sup> all of which are incompatible with the prohibition against aerial spraying under Ordinance No. 0309-07.

Although Memorandum Circular No. 02 and Ordinance No. 0309-07 both require the maintenance of the buffer zone, they differ as to their treatment and maintenance of the buffer zone. Under Memorandum Circular No. 02, a 50-meter "no-spray boundary" buffer zone should be observed by the spray pilots,<sup>193</sup> and the observance of the zone should be recorded in the Aerial Spray Final Report (ASFR) as a post-application safety measure.<sup>194</sup> On the other hand, Ordinance No. 0309-07 requires the maintenance of the 30-meter buffer zone to be planted with diversified trees.<sup>195</sup>

Devoid of the specific delegation to its local legislative body, the City of Davao exceeded its delegated authority to enact Ordinance No. 0309-07. Hence, Ordinance No. 0309-07 must be struck down also for being *an ultra vires* act on the part of the Sangguniang Bayan of Davao City.

We must emphasize that our ruling herein does not seek to deprive the LGUs their right to regulate activities within their jurisdiction. They are empowered under Section 16 of the Local Government Code to promote the general welfare of the people through regulatory, not prohibitive, ordinances that conform with the policy directions of the National Government. Ordinance No. 0309-07 failed to pass this test as it contravenes the specific regulatory policy on aerial spraying in banana plantations on a nationwide scale of the National Government, through the FPA.

Finally, the unconstitutionality of the ban renders nugatory Ordinance No. 0309-07 in its entirety. Consequently, any discussion on the lack of the separability clause becomes entirely irrelevant.

WHEREFORE, the Court DENIES the consolidated petitions for review on certiorari for their lack of merit; AFFIRMS the decision promulgated on January 9, 2009 in C.A.-G.R. CV No. 01389-MIN. declaring Ordinance No. 0309-07 UNCONSTITUTIONAL; PERMANENTLY ENJOINS respondent City of Davao, and all persons or entities acting in its behalf or under its authority, from enforcing and implementing Ordinance No. 0309-07; and ORDERS the petitioners to pay the costs of suit.

Sereno, C. J., Velasco, Jr., Leonardo-De Castro, Peralta, Del Castillo, Perez, Mendoza, Reyes, Perlas-Bernabe, Jardeleza, and Caguioa, JJ., concur.

Carpio, J., No part. Former law partners are counsels.

Leonen, J., See separate concurring opinion.

Please take notice that on August 16, 2016 a Decision/Resolution, copy attached herewith, was rendered by the Supreme Court in the above-entitled cases the original of which was received by this Office on September 9, 2016 at 10:15.

1 Rollo (G.R. No. 189185; Vol. I), pp. 72-115; penned by Associate Justice Jane Aurora C. Lantion, with the concurrence of Associate Justice Rodrigo F. Lim, Jr. (retired), Associate Justice Normandie B. Pizarro, and Associate Justice Michael P. Elbinias (deceased); while Associate Justice Romulo V. Borja dissented.

4 Rollo (G.R. No. 189185; Vol. I), p. 74.

5 Records no. 1, pp. 2-60; Entitled "Pilipino Banana Growers and Export Association, Inc., Davao Fruits Corporation and Lapanday Agricultural and Development Corporation, petitioners, versus City of Davao, respondent," docketed as Civil Case No. 31, 837-07.

6 Section 511. Posting and Publication of Ordinances with Penal Sanctions. (a) Ordinances with penal sanctions shall be posted at prominent places in the provincial capitol, city, municipal or barangay hall, as the case may be, for a minimum period of three (3) consecutive weeks. Such ordinances shall also be published in a newspaper of general circulation, where available, within the territorial jurisdiction of the local government unit concerned, except in the case of barangay ordinances. Unless otherwise provided therein, said ordinances shall take effect on the day following its publication, or at the end of the period, of posting, whichever occurs later.

(c) The secretary to the sanggunian concerned shall transmit official copies of such ordinances to the chief executive officer of the Official Gazette within seven (7) days following the approval of the said ordinance for publication purposes. The Official Gazette may publish ordinances with penal sanctions for archival and reference purposes.

7 Namely: Wilfredo Mosqueda, Marcelo Villaganes, Crispin Alcomendras, Corazon Sabinada, Rebecca Saligumba, Carolina Pilongo, Alejandra Bentoy, Ledevina Adlawan, and Virginia Cata-ag.

8 Namely: Geraldine Catalan, Julieta Lawagon and Florencia Sabandon.

10 Records no. 4, pp. 1115-1120.

11 Records no. 5, pp. 1422-1430, (The RTC issued the writ of preliminary injunction on June 25, 2007 after the PBGEA posted a P1,000,000.00 bond).

23 Rollo (G.R. No. 189185; Vol. I), pp. 209-227.

24 Rollo (G.R. No. 189195; Vol. I), pp. 39-42.

31 G.R. No. 104786, January 27, 1994, 229 SCRA 554, 559.

32 G.R. No. 156052, February 13, 2008, 545 SCRA 92, 142.

33 The ISO 14000 family of international standards provides practical management tools for companies and organizations in the management of environmental aspects and assessment of their environmental performance. (See International Organization for Standardization, "Environmental Management: The ISO 14000 family of International Standards," (4th ed., 2010) available at [www.iso.org/iso/home/store/publication\\_item.htm?pid=PUB100238](http://www.iso.org/iso/home/store/publication_item.htm?pid=PUB100238) last opened on July 14, 2016 at 9:00 a.m.)

34 Rollo (G.R. No. 189185; Vol. I), p. 62.

35 Rollo (G.R. No. 189305; Vol. I), pp. 82-83.

41Rollo(G.R. No. 189305; Vol. I), pp. 61-64.

46Section 6, Article XII, 1987 Constitution.

47Rollo(G.R. No. 189185; Vol. I), p. 375.

48Rollo(G.R. No. 189185; Vol. II), pp. 1244-1251.

49G.R. No. 118127, April 12, 2005, 455 SCRA 308, 342.

50Rollo(G.R. No. 189185; Vol. II), pp. 1265-1266.

51A period of four (4) to twelve (12) days.

52Rollo(G.R. No. 189185; Vol. II), pp. 1266-1267.

55Id. at 1257-1258; according to the respondents' witness, Mr. Richard Billington, the drift at the edge of an area sprayed from the air results to approximately half of the corresponding value for ground application. This observation was based on the AgDrift Model, developed under a Cooperative Research and Development Agreement (CRADA) between the Spray Drift Task Force (SDTF) of the US Environmental Protection Agency (EPA) and the US Department of Agriculture - Agricultural Research Service (USDA-ARS).

Category and Signal Words

Color Band Symbol

Acute Toxicity to Rat

Oral LD50 (mg/kg BW)

Dermal LD50 (mg/kg BW)

CATEGORY I DANGER:POISON

RED

50 or less

200 or less

100 or less

400 or less

CATEGORY II WARNING: HARMFUL

YELLOW

51 to 500

201 to 200

101 to 1000

401 to 4000

CATEGORY III CAUTION

BLUE

501 to 20000

2001 to 3000

Over 1000

Over 4000

CATEGORY IV

GREEN

Over 2000

Over 3000

N/A

N/A

FPA Classification Table of pesticides adopted from the World Health Organization (WHO) Classification by Hazards (RTC Records, No. 1, p. 41).

58According to the respondents' witness, Anacleto M. Pedrosa, Jr., Ph.D, acute toxicity to rats of Category IV fungicides require oral ingestion of over 2000 milligrams in solid form per kilogram of body Weight and over 3000 milligrams of such fungicide in liquid form per kilogram of body weight to have any Adverse effect (See RTC Records, No. 4, pp. 1095-1096.)

59Rollo(G.R. No. 189185; Vol. III), pp. 1545-1554; Entitled "Summary Report on the Assessment and Finding Activities on the Issue of Aerial Spraying in Banana Plantations".

60Rollo(G.R No. 189185; Vol. II), pp. 1271-1273.

65"Good agricultural practice" is broadly defined as applying knowledge to addressing environmental, economic and social sustainability for on-farm production and post-production processes resulting in safe and healthy food and non-food agricultural products. The use of pesticides includes the officially recommended or nationally authorized uses of pesticides under actual conditions necessary for effective and reliable pest control. It encompasses a range of levels of pesticide applications up to the highest authorized use, applied in a manner that leaves a residue which is the smallest amount practicable. See FAO-Committee on Agriculture, "Development of a Framework for Good Agricultural Practices" (Rome. March 31-April 4, 2003),<http://www.fao.org/docrep/meeting/006/y8704e.html>last accessed July 14, 2016 at 9:40 a.m.

66The Guide offers practical help and guidance to individuals and entities involved in using pesticides for food and fibre production as well as in Public Health programmes. They cover the main terrestrial and aerial spray application techniques. The guide also identifies some of the problems and suggest means of addressing them. See FAO-Committee on Agriculture and Consumer Protection, "Guidelines on Good Practice for Aerial Application of Pesticides (Rome, 2001),<http://www.fao.org/docrep/006/y2766e/y2766e00.html>last accessed July 14, 2016 at 9:42 a.m.

67Rollo(G.R. No. 189185; Vol. II), pp. 1300-1301; this includes: (a) notice to the community through the advisory board at least three (3) days before the scheduled date of spraying; (b) determining the flight pattern for the aircraft applicator using the Differential Global Positioning system (DGPS) to establish precise swath patterns and determine specific points during the flight for the spray valve to be turned on and shut off; (c) pre-flight inspection of the aircraft, including the cleaning and checking of the spray valves in the Micronair Rotary Drift Control Atomizers (AU 5000 Low-Drift model) that disperses the solution being sprayed for a consistent droplet-size of 200 to 250 microns to control drift; (d) monitoring by the Spray Supervisor of the weather and environmental conditions in the weather station; and (e) sounding of alarms for fifteen (15) minutes prior to take-off.

68Id. at 1301; the following are observed: (a) monitoring of wind speed and direction, and weather conditions, and maintaining radio contact with the pilot during aerial spraying operations; (b) diverting road traffic to prevent people from traversing areas near the plantations; (c) maintaining a flying height clearance of about 3.5 meters above the leaf canopy; (d) ensuring that spraying valves are shut-off at least 50 meters before the edge of the perimeter and before the 30 meter buffer zone.

69Id. at 1302; includes: (a) DGPS data card recording the swath pattern submitted to the Spray Supervisor; and (b) cleaning of aircraft including the Micronair Rotary Drift Control Atomizers which is being calibrated monthly.

70Id. at 1302-1303; respondents allegedly invested in sensors, wind meters, wind cones, field thermometers and a central weather station.

71Id. at 1330; A precision satellite-based navigational system that accurately plots the plantation and guides the pilot in conducting aerial spraying.

72Id.; An instrument that depicts an accurate map of the plantation, indicating the turn-on and shut-off spray valve points during the flight, and records swath patterns while the aerial spraying is being conducted.

73Id.; Allows the pilot to program the grid coordinates of a particular plantation on the DGPS, retrieve navigational guidance for the pilot, monitor ground speed (tailwind and headwind), program and retrieve date to record the actual spraying operation.

74Id.; Ensures that the droplets of solution released for aerial spraying are consistently delivered with each droplet with a size of 250 microns to control drift. It controls the flow and the drift of the solution released for aerial spraying even when the aircraft applicator is operating at 145-240 kilometers per hour.

75Id.; Controls the rate of application of the solution for aerial application to ensure that the substance being aerially sprayed is consistently and equally applied throughout the entire banana plantation.

76Id.; A device that will automatically turn on and shut off the spray valves on precise points within the target area as programmed in the GPS.

77Rollo(G.R. No. 189185; Vol. II), p. 1331.

85Philippine Center for Postharvest Development and Mechanization (PhilMech), "Banana Post-harvest Situationer,"<http://www.philmech.gov.ph/phindustry/banana.htm>., last accessed July 14, 2016 at 9:44 a.m.

86DA High Value Crops Development Program,<http://hvcc.da.gov.ph/banana.htm>, last accessed July 14, 2016 at 9:46 a.m.

87Philippine Statistics Authority, "Philippine Agriculture in Figures, 2013,"<http://countrystat.psa.gov.ph/?cont=3>, last accessed July 14, 2016 at 9:50 a.m.

88Includes Davao del Norte, Davao City, Compostela Valley, Davao Oriental and Davao del Sur, Panabo City, Tagum, Digos, Island Garden City of Samal.

89Philippine Statistics Authority, "Regional Profile: Davao,"<http://countrystat.psa.gov.ph/?cont=16&r=II>, last accessed July 14, 2016 at 9:55 a.m.

90Farms infested by Panama disease are abandoned and left idle for about five years before recultivation. In Davao City, only 1,800 hectares of the original 5,200 hectares planted to bananas have remained due to the infection. (<http://www.ugnayan.com/ph/DavaodelSur/Davao/article/YCL>, last accessed April 4, 2015 at 1:57 p.m.) Only two (2) varieties of Cavendish banana are recommended for planting in Effected soil. Otherwise, new crops such as com, cacao and oil palm are recommended for cultivation. See Manuel Cayon, "DA allots P102 million for Panama-disease control among banana growers: Business Mirror (28 April 2015),[www.businessmirror.com.ph/2015/04/28/da-allots-p102million-for-panama-disease-control-among-banana-growers](http://www.businessmirror.com.ph/2015/04/28/da-allots-p102million-for-panama-disease-control-among-banana-growers)).

91Ploetz, Randy, "Black Sigatoka of Banana: The Most Important Disease of a Most Important Fruit," APS, 2001,<http://www.apsnet.org/publications/apsnetfeatures/Pages/blacksigatoka.aspx>, last accessed July 14, 2016 at 10:08 a.m.

92<https://www.wageningenur.nl/en/show/Another-malor-step-in-better-disease-manasement-in-the-global-banana-sector.htm>, last accessed July 14, 2016 at 10:11 a.m.

93Banana: Diseases,<http://nhb.gov.in/fruits/banana/ban002.pdf>, last accessed July 14, 2016 at 10:15 a.m.

94Ploetz, Randy, Black Sigatoka in Pesticide Outlook, Vol. 11, Issue 2000,[www.researchinformation.co.uk/pest/2000/B006308H/pdf](http://www.researchinformation.co.uk/pest/2000/B006308H/pdf), last accessed July 14, 2016 at 10:21 a.m.

95Ploetz, Randy, "Black Sigatoka of Banana: The Most Important Disease of a Most Important Fruit," APS, 2001,<http://www.apsnet.org/publications/apsnetfeatures/Pages/blacksigatoka.aspx>, last accessed July 14, 2016 at 10:13 a.m.

96Rollo(G.R. No. 189185; Vol. III), p. 1548; Summary Report on the Assessment and Factfinding Activities on the Issue of Aerial Spraying in Banana Plantations.

97Id. at 1547; Summary Report on the Assessment and Factfinding Activities on the Issue of Aerial Spraying in Banana Plantations.

98Id. at 1549; Summary-Report on the Assessment and Factfinding Activities on the Issue of Aerial Spraying in Banana Plantations.

100Legaspi v. City of Cebu, G.R. No. 159110, December 10, 2013, 711 SCRA 771, 785.

101Section 53. Quorum. - (a) A majority of all the members of the sanggunian who have been elected and qualified shall constitute a quorum to transact official business. x x x

102Section 54. Approval of Ordinances. - (a) Every ordinance enacted by the x x x sangguniang panlungsod x x x shall be presented to the x x x city or municipal mayor, as the case may be. If the local chief executive concerned approves the same, he shall affix his signature on each and every page thereof; x x x.

- 103Social Justice Society (SJS) v. Atienza, Jr., G.R. No. 156502, 13 February 2008, 545 SCRA 92, 139-140.
- 104Rural Bank of Makati, Inc. v. Municipality of Makati, G.R. No. 150763, July 2, 2004, 433 SCRA 362, 371.
- 105Sec. 458, Article III, Title III, Book III, R.A. No. 7160.
- 106Rural Bank of Makati, Inc. v. Municipality of Makati, G.R. No. 150763, July 2, 2004, 433 SCRA 362, 371-372;United States v. Salaveria, 39 Phil. 102, 110 (1918).
- 107G.R. No. 101083, July 30, 1993, 224 SCRA 792, 805.
- 108Fernando v. St. Scholastica's College, G.R. No. 161107, March 12, 2013, 693 SCRA 141, 157, citingWhite Light Corporation v. City of Manila, No. G.R. No. 122846, January 20, 2009, 576 SCRA 416, 433.
- 109Legaspi v. City of Cebu, G.R. No. 159110, December 10, 2013, 711 SCRA 771, 784-785; citingCity of Manila v. Laguio, Jr., G.R. No. 118127, April 12, 2005, 455 SCRA 308, 326.
- 112Parayno v. Jovellanos, G.R. No. 148408, July 14, 2006, 495 SCRA 85, 93.
- 113City of Manila v. Laguio, Jr., G.R. No. 118127, April 12, 2005, 455 SCRA 308, 330.
- 114State v. Old South Amusements, Inc., 564 S.E.2d 710 (2002).
- 115See RTC Decision, RTC records No. 10, pp. 2926-2927.
- 116Rollo(G.R. No. 189185; Vol. I), pp. 86-91.
- 117Id. at 1542-2543; based on the report submitted by Engr. Magno Porticos, Jr., the cost and time frame estimate submitted to the PBGEA was based on the requirements of lowland and relatively flat lands where road and drainage system to be constructed will be uniformly straight and equidistant. The cost for plantations consisting of slope terrains and gullies, will vary. See Engineering Committee Report on the Main Engineering Works Needed to Comply with the Ordinance Banning Aerial Spray.
- 118G.R. No. 118127, April 12, 2005, 455 SCRA 308, 339-342.
- 119Biraogo v. The Philippine Truth Commission of 2010, G.R. No. 192935, December 7, 2010, 637 SCRA 78, 167.
- 120Bartolome v. Social Security System, G.R. No. 192531, November 12, 2014;Garcia v. Executive Secretary, G.R. No. 198554, July 30, 2012, 677 SCRA 750, 177.
- 121JMM Promotion and Management, Inc. v. Court of Appeals, G.R. No. 120095, August 5, 1996, 260 SCRA 319, 331.
- 122Quinto v. Commission on Elections, G.R. No. 189698, February 22, 2010, 606 SCRA 258, 414.
- 123Tiu v. Court of Appeals, G.R. No. 127410, January 20, 1999, 301 SCRA 278, 288.
- 124City of Manila v. Laguio, Jr., G.R. No. 118127, April 12, 2005, 455 SCRA 308, 348-349.
- 125Russell W. Galloway, "Means-End Scrutiny in American Constitutional Law" Loyola of Los Angeles Law Review, Vol. 21, p. 449, available at<http://digitalcommons.lmu.edu/cgi/viewcontent.cgi?article=1557&context=llr>last accessed August 16, 2016.
- 126Social Justice Society (SJS) v. Atienza, Jr., G.R. No. 156052, February 13, 2008, 455 SCRA 92, 138.
- 127See the Concurring Opinion of Justice Teresita J. de Castro inGarcia v. Drilon, G.R. No. 179267, June 25, 2013, 699 SCRA 435, 447.
- 128Graham v. Richardson, 403 U.S. 365, 91 S.Ct. 1848, 29 L.Ed.2d 532 (1971).
- 129Id. Suspect class refers to alienage such as that based on nationality or race.
- 130Marcy Strauss,Reevaluating Suspect Classifications, Seattle University Law Review, Vol. 35:135, p. 146, available at<http://digitalcommons.law.seattleu.edu/cgi/viewcontent.cgi?article=2059&context=sulr>, last accessed August 16, 2016;White Light Corporation v. City of Manila, G.R. No. 122846, January 20, 2009, 576 SCRA 416, 436-437.
- 131See Separate Concurring Opinion of J. Puno (ret.) inAng Ladlad LGBT Party v. Commission on Elections, G.R. No. 190582, April 8, 2010, 618 SCRA 81, 94.

132Disini, Jr. v. Secretary of Justice, G.R. No. 203335, February 18, 2014, 716 SCRA 237, 301.

133In determining the reasonableness of a classification, one must look beyond the classification to the purpose of the law which is the elimination of a mischief. This gives rise to two (2) classes: the first consists of all individuals possessing the defining character or characteristics of the legislative classification ("Trait"); the second would consist of all individuals possessing or tainted by the mischief at which the law aims. See Joseph Tussman and Jacobus tenBroek, *The Equal Protection of the Laws*, 37 CAL. L. REV. 341 (1949), available at <http://scholarship.law.berkeley.edu/cgi/viewcontent.cgi?article=3493&context=californialawreview> Last accessed August 16, 2016.

134RTC records no. 8, pp. 2361-2362 (Submitted as Exhibit "10" of the petitioners-intervenors).

135This includes Hand sprayers and atomizers, Hand compressed sprayers, Knapsack sprayers, Tractor-mounted sprayer, Motorized knapsack mist blowers, Ultra low volume or controlled-droplet applicators (ULV/CDA), Fogging machines/fogair sprayers, Hand-carried dusters, Hand-carried granule applicators, Power dusters, Aerial application (Aircraft sprayers), and Injectors and fumigation equipment (S.K. Pal and S.K. Das Gupta, "Pesticide Application" Skill Development Series-No. 17, ICRISAT Training and Fellowship Program, International Crops Research Institute for the Semi-Arid Tropics, available at <http://oar.icrisat.org/2430//Pesticide-Application.pdf>, accessed August 16, 2016, 1:52 p.m.

136Food and Agriculture Organization of the United States. *Guidelines on Good Agricultural Practice for Ground Application of Pesticides*. Rome 2001.

137Susan Cordell, and Paul B. Baker, *Pesticide Drift*, available at <http://extension.arizona.edu/sites/extension.arizona.edu/files/pubs/az1050.pdf>, last accessed August 16, 2016.

139F.M. Fishel and J.A. Ferrell, *Managing Pesticide Drift*, available at <http://edis.ifas.ufl.edu/pi232>, last accessed August 16, 2016.

140Rollo (G.R. No. 189185; Vol. III), p. 1548; Summary Report on the Assessment and Factfinding Activities on the Issue of Aerial Spraying in Banana Plantations.

141Rollo (G.R. No. 189185; Vol. III), p. 1549; Summary Report on the Assessment and Factfinding Activities on the Issue of Aerial Spraying in Banana Plantations.

142Id. at 1566; According to Regional Health Director of the Department of Health (DOH) Paulyn Jean B. Rosell-Ubial (now the Secretary of Health), the ban against aerial spraying and adoption of ground spraying would not eliminate the hazards of the pesticides to which workers and residents within and around banana plantations might be exposed.

144David M. Treiman, *Equal Protection and Fundamental Rights - A Judicial Shell Game*, 15 Tulsa L. J. 183, 191 (1979), available at: <http://digitalcommons.law.utulsa.edu/cgi/viewcontent.cgi?article=1510&context=tlr>, last accessed August 16, 2016.

145Rollo, (G.R. No. 189185; Vol. II), pp. 1257-1258; According to respondents' witness, Mr. Richard Billington, the drift at the edge of an area sprayed from the air results to approximately half of the corresponding value for ground application. This observation was based on the AgDrift Model, developed under a Cooperative Research and Development Agreement (CRADA) between the Spray Drift Task Force (SDTF) of the US Environmental Protection Agency (EPA) and the US Department of Agriculture - Agricultural Research Service (USDA-ARS).

146Tussman and tenBroek, *supra* at 133.

147Rollo, G.R. No. 189185, Vol. I, pp. 102-103.

148The rational basis approach partakes of two (2) forms: the deferential and then nondeferential rational relation test. In deferential rational basis test, the government action is always deemed constitutional if it has any conceivable valid purpose and if the means chosen are arguably rational. In contrast, the nondeferential rational basis test requires a determination that the government action serves an actual valid interest, hence (1) the government actually has a valid purpose and (2) the means chosen are demonstrably rational (effective), see Galloway, Russell W., *Means-End Scrutiny in American Constitutional Law*, Loyola of Los Angeles Review, Vol. 21, pp. 451-452, available at <http://digitalcommons.lmu.edu/cgi/viewcontent.cgi?article=1557&context=llr>, last accessed August 16, 2016.

150Cabell v. Chavez-Salido, 454 U.S. 432 (1982), 70 L.Ed.2d 677.

152Rollo(G.R. No. 189185; Vol. III), pp. 1564-1565.

154Andrew Jordan and Timothy O'Riordan, "The Precautionary Principle: A Legal and Policy History" in The precautionary principle: Protecting Public Health, The Environment and The Future of Our Children, p. 33, available at[http://www.euro.who.int/\\_data/assets/pdf\\_file/0003/91173/E83079.pdf](http://www.euro.who.int/_data/assets/pdf_file/0003/91173/E83079.pdf), last accessed August 16, 2016.

155UNESCO. The Precautionary Principle, World Commission on the Ethics of Scientific Knowledge and Technology (COMEST), p. 7, available at<http://www.eubios.info/UNESCO/precprin.pdf>, last accessed August 16, 2016.

157Annotation to the Rules of Procedure on Environmental Cases, p. 158.

158IUCN,Guidelines for Applying the Precautionary Principle to Biodiversity Conservation and Natural Resource Management, available at[http://www.cmsdata.iucn.org/downloads/In250507\\_ppguidelines.pdf](http://www.cmsdata.iucn.org/downloads/In250507_ppguidelines.pdf).Last accessed August 16, 2016.

160Andrew Stirling and Joel Tickner, "Implementing Precaution: Assessment and Application Tools for Health and Environmental Decision-Making" in The Precautionary Principle: Protecting Public Health, The Environment and The Future of Our Children, p. 182, available at[http://www.euro.who.int/\\_data/assets/pdf\\_file/0003/91173/E83079.pdf](http://www.euro.who.int/_data/assets/pdf_file/0003/91173/E83079.pdf)Last accessed August 16, 2016.

162European Commission Communication from the Commission on the Precautionary Principle, available at<http://eur-lex.europa.eu/legal-content/EN/TXT/?uri=URISERV%3A132042>. Last accessed August 16, 2016.

165Position Paper of the Department of Health-Center for Health Development, Davao Region, On the Issue of Aerial Spraying in Banana Plantations Within the Jurisdiction of Davao City in G.R. No. 189185, Vol. III, pp. 1566-1567.

166United States v. Salaveria, 39 Phil. 102, 111 (1918).

168G.R. No. 177807, October 11, 2011, 658 SCRA 853, 865-866.

171De la Cruz v. Paras, G.R. No. L-42571-72, July 25, 1983, 123 SCRA 569, 578.

172City of Manila v. Laguio, Jr., G.R. No. 118127, April 12, 2005, 455 SCRA 308, 353.

173White Light Corporation v. City of Manila, G.R. No. 122846, January 20, 2009, 576 SCRA 417, 442 citingMorfe v. Mutuc, No. L-20387, January 31, 1968, 22 SCRA 424, 441.

174City of Manila v. Laguio, Jr., G.R. No. 118127, April 12, 2005, 455 SCRA 308, 327.

175Legaspi v. City of Cebu, G.R. No. 159692, December 10, 2013, 711 SCRA 771, 785.

176Creating the Fertilizer and Pesticide Authority and Abolishing the Fertilizer Industry Authority.

179The delivery of basic services is devolved to the local government units. Sections 22 and 458 of the Local Government Code provides for an exhaustive enumeration of the functions and duties devolved to the local government units.

180Batangas CATV v. Court of Appeals, G.R. No. 138810, September 29, 2004, 439 SCRA 326, 340.

183G.R. No. 118127, April 12, 2005, 455 SCRA 308, 327.

186The memorandum provides for the safety procedures in pesticide spraying, (Paragraph. II [1]), safety handling (Paragraph II [2]) and post-application (Paragraph II [3]), including the qualification of applicators ((Paragraph III), storing of fungicides (Paragraph IV), safety and equipment of plantation personnel (Paragraph V).

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